

AI Summaries Overrepresent Fake Reviews: Evidence from Amazon

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Abstract

Digital platforms increasingly deploy AI-generated review summaries to help consumers navigate large volumes of online reviews. We document an unintended consequence of AI summaries: they overrepresent fake reviews. Because AI summarization algorithms are designed to capture recurring topics across texts, they naturally capture the generic praise characteristic of fake reviews. Using data from Amazon, we provide both review-level and product-level evidence of this overrepresentation. Moreover, we find that products with greater exposure to review manipulation receive more positive AI summaries, even after controlling for average ratings and other observable product characteristics. Finally, we show that products with greater exposure to review manipulation experience relative improvements in sales ranks following the introduction of AI summaries. These findings demonstrate that AI systems designed to mitigate information overload may inadvertently amplify misleading information when operating in a contaminated information environment.

Keywords: AI summaries, AI bias, fake reviews, digital platforms, misinformation

1 Introduction

Digital platforms, such as Amazon, Best Buy, and the Apple App Store, increasingly use AI-generated summaries of reviews to help consumers digest online reviews. [Schermerhorn \(2023\)](#), the director of community shopping at Amazon, describes the goal of this feature as helping consumers more easily grasp common topics from the vast volume of reviews on the platform. AI review summarization typically combines topic extraction and sentiment analysis: it identifies recurring topics in the review corpus and aggregates the sentiment of reviews that mention each topic. On Amazon, this procedure generates a list of keywords assigned to extracted topics, the sentiment labels that display the aggregated sentiment associated with each extracted topic (either positive, neutral or negative), and a short paragraph that is organized around the extracted topics and their corresponding sentiment.

Although this procedure seems intuitive, it is important to note that not all reviews are treated equally in this process. Specifically, reviews that mention more extracted topics carry higher weights in AI summaries. This selective representation raises concerns because consumer review corpora often include fake reviews ([Mayzlin et al., 2014](#)). Indeed, major news organizations such as *The Wall Street Journal* and the *Associated Press* continue to warn that fake reviews remain widespread ([Nguyen, 2021](#); [Hadero, 2024](#)).

Against this backdrop, we argue that AI summaries unintentionally overrepresent fake reviews. This occurs because fake reviews tend to emphasize generic product merits rather than idiosyncratic experiences, and such generic merits align with common topics that AI summarization algorithms are designed to extract. Furthermore, because fake reviews are typically more positive in tone than authentic reviews, their overrepresentation may make AI summaries for review manipulators’ products overly positive, even after conditioning on average ratings, and thereby increase those products’ sales.

Empirically testing whether fake reviews are overrepresented poses key challenges, because researchers rarely observe either which reviews are fake or which extracted topics a given review mentions. We address these obstacles through complementary review-level and product-level analyses. First, at the review level, we leverage a publicly available dataset with pre-classified fake and authentic reviews. We apply lexical and semantic matching to determine the number of

extracted topics mentioned in each review. Across different specifications, we consistently find that fake reviews mention significantly more extracted topics than authentic reviews, providing direct evidence of the overrepresentation of fake reviews. Second, at the product level, we extend the analysis to a broader sample of products that appear in Amazon search results. For each product, we use information publicly available on Amazon’s interface to calculate the average number of extracted topics mentioned per review. We find that this measure is higher among products with greater exposure to fake reviews, using non-Amazon-sold status and low review-credibility grades from RateBud as complementary proxies. This provides further empirical support for the overrepresentation of fake reviews using the actual outputs of Amazon’s deployed AI summarization algorithms.

In addition, we use OpenAI GPT-4.1 and a specialized fine-tuned BERT model to analyze the sentiment of AI summaries across products; we find that products with greater exposure to fake reviews receive systematically more positive sentiment in both topic labels and summary paragraphs, even after controlling for average ratings and other observable characteristics.

Finally, we investigate whether this algorithmic overrepresentation generates market distortions. We find that after the introduction of AI summaries, Amazon-sold products experience a deterioration in sales ranks relative to their competitors, whereas products assigned low-review-credibility grades by RateBud experience a relative improvement. Because our design absorbs pre-existing sales differences, including those potentially attributable to fake reviews through channels other than AI summaries, these findings suggest that AI summaries may amplify the market impact of fake reviews.

Our results point to a governance challenge rather than merely a technical flaw. As [Cowgill and Tucker \(2020\)](#) emphasize, bias rooted in algorithmic objectives cannot be remedied merely through technical fixes. In our setting, the algorithm can accurately identify the most prevalent topics and nevertheless systematically overrepresent fake reviews, because the objective it optimizes is unintentionally aligned with the textual features of fake reviews. This bias therefore does not require a prediction error and need not disappear when predictive accuracy improves; in fact, improving the algorithm’s ability to optimize its existing objective could further intensify the bias. For platform managers, our findings highlight the need to look beyond the immediate convenience of AI summaries and evaluate whether the systematic overrepresentation of fake reviews creates

longer-term risks to consumer welfare, user trust, and overall platform integrity. For policymakers, effective regulation must address not only the creation of fake reviews but also the algorithmic systems that aggregate and amplify them. More broadly, the problem calls for greater societal attention and the development of next-generation algorithms that remain robust in strategically manipulated information environments.

The remainder of the paper is organized as follows. Section 2 reviews the literature and elaborates on our contributions to the literature. Section 3 introduces the institutional setting and data. Section 4 develops a theoretical framework that defines the overrepresentation of fake reviews and proposes hypotheses. Section 5 presents the empirical findings. Section 6 concludes with the implications of the findings.

2 Literature Review

Our work intersects four streams of literature: AI bias, fake reviews, the spread of misinformation, and AI-generated summaries. The literature on AI and algorithmic bias has attracted widespread interdisciplinary attention. Cowgill and Tucker (2020) distinguish between algorithmic bias arising from an algorithm’s objectives (e.g., Levy, 2021; Nyhan et al., 2023; Lambrecht and Tucker, 2019) and bias stemming from its biased predictions (e.g., Blodgett and O’Connor, 2017). They also highlight the particular challenges of dealing with objective-induced bias. We contribute by documenting objective-induced bias in AI summarization: AI summarization algorithms designed to identify common topics can systematically overrepresent fake reviews.

We also contribute to the literature on fake reviews. Mayzlin et al. (2014) provide early empirical evidence of the existence of fake reviews, and Luca and Zervas (2016) study the economic incentives underlying review manipulation. He et al. (2022a) and He et al. (2022b) contribute significantly to the literature by directly observing the market for fake reviews, thereby enabling more confident identification of fake reviews and providing suggestive evidence of their negative welfare effects. Gandhi et al. (2025) further study the welfare implications of fake reviews. Our paper contributes to this stream of literature by suggesting that the negative welfare impact of fake reviews might be reinforced by AI summaries in the AI era.

A third related stream of research examines the spread of misinformation. [Vosoughi et al. \(2018\)](#) provide strong empirical evidence that misinformation diffuses significantly faster, farther, and deeper than authentic information. Empirical studies further show that misinformation diffusion can be facilitated by social bots ([Shao et al., 2018](#)), people’s tendency to engage more with misinformation ([Vosoughi et al., 2018](#)), people’s failure to consider the accuracy of information ([Pennycook et al., 2021](#)), overconfidence in one’s ability to spot misinformation ([Lyons et al., 2021](#)), political polarization ([Zhu and Pechmann, 2024](#)), and echo chambers ([Del Vicario et al., 2016](#)). Our paper contributes to this stream of literature by showing that AI can also amplify the voice of fake reviews, which may contain misleading and inaccurate information.

Lastly, [Wang et al. \(2025\)](#) and [Wang and Wang \(2025\)](#) find that AI summaries increase exploration and purchase rates. Ongoing research by [Su et al. \(2025\)](#) finds that the introduction of AI summaries leads subsequent reviewers to express more distinctive perspectives than earlier reviewers. To our knowledge, this is the first study to document that AI summaries overrepresent fake reviews.

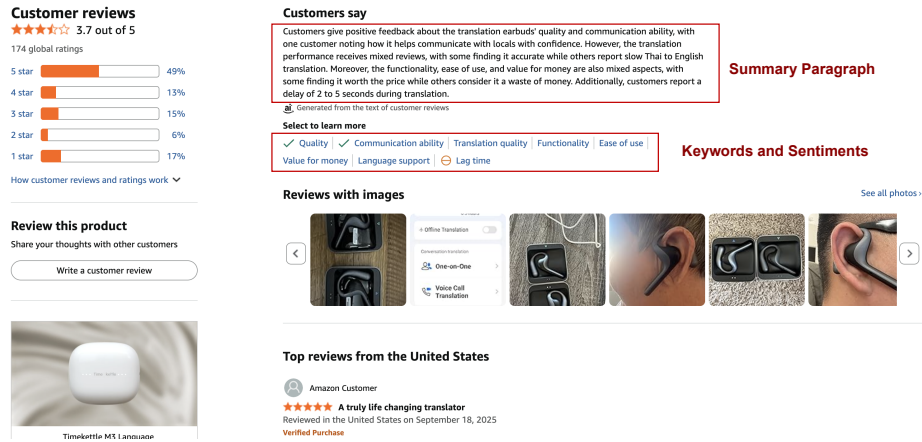
3 Setting and Data

3.1 AI Review Summaries

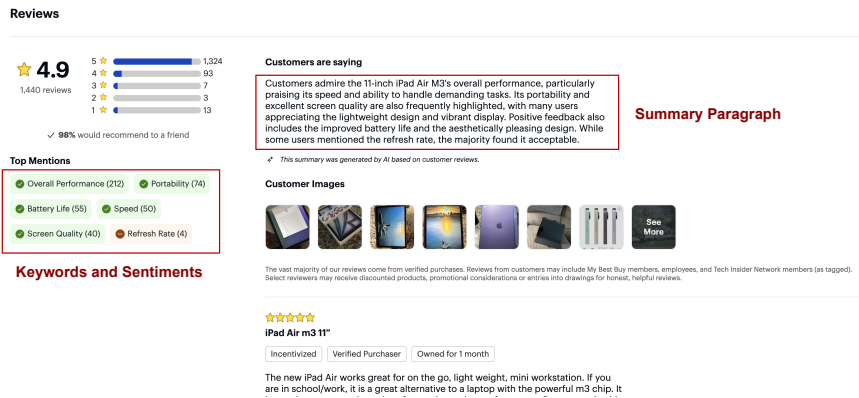
AI summaries are generally displayed in a prominent location above all user-generated reviews. Panels (a) and (b) of Figure 1 illustrate how AI summaries are displayed on Amazon and Best Buy. AI summaries typically consist of two user-facing components: (1) a list of keywords, each corresponding to an extracted topic and accompanied by a topic-level sentiment label, and (2) a summary paragraph. Throughout the remainder of the paper, we use “keyword” and “extracted topic” interchangeably. While some platforms feature both components (e.g., Amazon and Best Buy), others display only keywords with topic-level sentiment labels (e.g., Yelp) or standalone summary paragraphs (e.g., Apple App Store). When both components are present, the summary paragraph is typically structured around the extracted topics and their corresponding sentiment.

In this paper, we focus on Amazon, which is an important setting for studying review summaries due to its popularity—more than half (56%) of consumers in the US start their shopping searches on Amazon ([Lebow, 2024](#)). Regarding the business goal of AI summaries, [Schermerhorn \(2023\)](#), the director of community shopping at Amazon, states that Amazon aims “to make it easier for

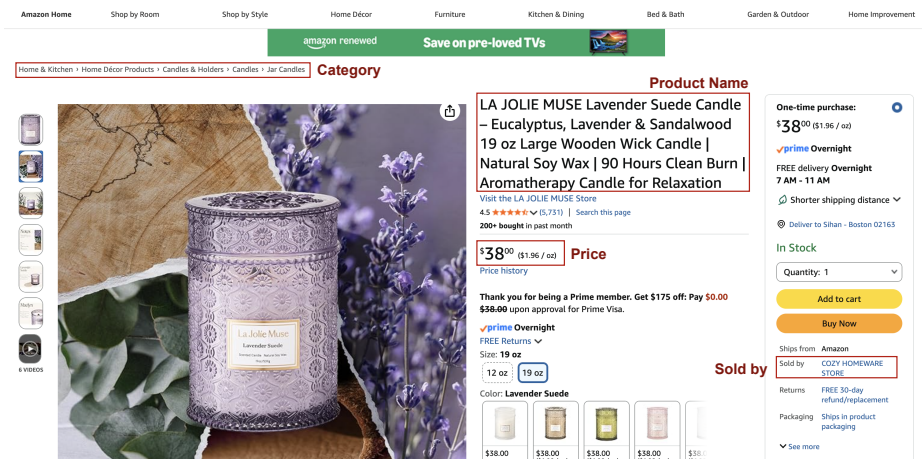
Figure 1: AI Summary Interfaces and Amazon Product Information



(a) Amazon AI Summary Interface



(b) Best Buy AI Summary Interface



(c) Locations of Product Information on Amazon

Note. Panels (a) and (b) were captured on November 1, 2025, from <https://www.amazon.com/dp/B0DWJCK9LZ/> and <https://www.bestbuy.com/product/apple-11-inch-ipad-air-m3-chip-built-for-apple-intelligence-wi-fi-128gb-blue/JJGCQ8VZVZ>, respectively. Panel (c) was captured on March 3, 2026, from <https://www.amazon.com/dp/B0B74R1VKS/>.

customers to understand the common themes across reviews. ... to help customers determine at a glance whether a product is right for them.” Amazon announced the introduction of the feature on August 14, 2023 (Schmerhorn, 2023).¹ Over time, the AI summary feature was gradually rolled out to a growing number of products (Su et al., 2025).

Gozluklu et al. (2025), members of Amazon’s internal AI team, explain that Amazon’s AI summarization algorithm first applies traditional machine-learning models to extract the most common topics from user-generated reviews and assign keywords to those topics. Sentiment analysis is then conducted at the topic level using reviews that mention each extracted topic. The resulting review-level sentiment classifications are aggregated into topic-level sentiment labels such as positive, neutral, or negative. Finally, Amazon uses small language models to generate the summary paragraphs.² This structure is important to our argument. Because AI summaries are organized around extracted common topics, reviews that mention more extracted topics receive greater representation in the AI summaries; reviews that do not mention them receive little or no representation.

3.2 Fake Reviews

Review manipulation on Amazon typically operates through off-platform coordination. Third-party sellers who seek to manipulate reviews create private groups on social media platforms such as Facebook, Twitter, and Telegram to recruit and communicate with fake reviewers (Stieb, 2022). These manipulators offer consumers full product refunds, and occasionally additional cash compensation, in exchange for writing five-star reviews (Huang et al., 2025).

The economics of off-platform reimbursement suggests that we should see little to no fake-review activity aimed at sabotaging competitors. Because most product categories on Amazon feature many competing sellers, a seller would need to purchase a prohibitively large number of negative reviews targeting numerous competitors to gain a meaningful competitive advantage. Indeed, He et al. (2022b) find no evidence that sellers purchase fake reviews to damage competitors’ reputations. They also find no evidence of review manipulation for Amazon-sold (first-party)

¹It is worth pointing out that media first reported the introduction of AI summaries in June 2023 (Palmer, 2023; Cerullo, 2023).

²Apple deploys a similar AI summarization system in the App Store. See <https://machinelearning.apple.com/research/app-store-review>, accessed on February 21, 2026.

products, likely because the severe reputational loss resulting from public exposure would far outweigh gains from fake reviews.

In terms of textual features of fake reviews, [Ott et al. \(2011\)](#) find that fake reviews contain more superlatives, use more positive and fewer negative emotion terms, and include less concrete language than authentic reviews. [Li et al. \(2014\)](#) show that fake reviews exhibit excessively positive sentiment. [Luca and Zervas \(2016\)](#) document similar patterns among suspicious reviews filtered out by Yelp. Prior research also shows that many fake reviews are highly similar to each other (e.g., [Jindal and Liu, 2008](#); [Mukherjee et al., 2012](#); [Rayana and Akoglu, 2015](#); [Wang et al., 2023](#); [Gupta et al., 2024](#)), and that fake reviewers tend to work collaboratively in groups ([Rathore et al., 2021](#)). Taken together, these findings suggest that fake reviews tend to be more positive, less concrete, and more mutually similar than authentic reviews.

These textual patterns are consistent with the incentives in the market for fake reviews. Because fake reviewers are often asked to write favorable comments about products within a short period of time, they may not have the opportunity to thoroughly test the products; consequently, they rely more on generic product merits than on specific usage details. Authentic reviewers, by contrast, may describe more idiosyncratic details based on their actual consumption experiences. This difference is central to the mechanism detailed in [Section 4.2](#).

3.3 Data

Our empirical analysis relies on four data sources. Data Source 1 is a publicly available dataset compiled by Brett Hollenbeck from several prior studies ([He et al., 2022a](#); [Feldman et al., 2025](#)) and is available in Hollenbeck’s GitHub repository ([Hollenbeck, 2026](#)). The dataset was collected in 2019–2020, and it contains products classified as fake review buyers by [He et al. \(2022a\)](#) and fake reviewers classified by [Feldman et al. \(2025\)](#).³ A review is classified as “fake” if it satisfies three criteria: (i) a five-star review, (ii) written for a review-manipulating seller identified by [He et al. \(2022a\)](#), and (iii) written by a fake reviewer identified by [Feldman et al. \(2025\)](#); otherwise, a review is classified as “authentic.” Some products in the dataset contain only authentic reviews; we exclude these products from our analyses. The product-level summary statistics are reported in

³[He et al. \(2022a\)](#) and [Feldman et al. \(2025\)](#) both build on directly observed fake review activity to construct high-confidence labels. They then use these labels to train classifiers that extrapolate identification to unlabeled observations.

the top panel of Table 1. The average number of reviews per product is around 110; this number aligns with Amazon’s interface limit, which restricts users to viewing ten pages of reviews per product (roughly 100 reviews in total, at about 10 reviews per page). The average percentage of fake reviews per product is around 39%; this seems high but it is expected because the dataset is constructed from products of review manipulators, rather than representing the overall population of Amazon products.

Table 1: Summary Statistics for Data Sources 1 (Fake Reviews), 2 (Amazon), and 4 (Keepa)

Statistic	Mean	SD	25th pct.	Median	75th pct.
<i>Data Source 1: Hollenbeck (2026); product-level summary; sample size = 1,459 products</i>					
# Reviews	109.89	188.78	18	46	116
Percentage of Fake Reviews	39.26%	26.24%	17.46%	36.36%	60.00%
<i>Data Source 2: Amazon; product-level summary; products with AI summaries; sample size = 14,479 products</i>					
# Reviews	7,879.91	26,287.32	296	1,307	5,453
Average Rating	4.436	0.276	4.3	4.5	4.6
Price	132.20	469.90	19.99	39.95	119.98
Percentage of Amazon-Sold Products	32.16%				
<i>Data Source 4: Keepa; product-day panel; sample size = 6,141,891 product-day observations from 6,823 products over 1,049 consecutive dates</i>					
Sales Rank	34,036.66	119,316.82	1,118	5,616	24,295
Log(Sales Rank)	8.46	2.28	7.02	8.63	10.10

Table 2: Review-Level Summary Statistics of Data Source 1

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Length (# Characters)	302.30	230.48 ($p < 0.001$)	244.64 ($p < 0.001$)
Length (# Words)	57.66	46.02 ($p < 0.001$)	43.43 ($p < 0.001$)
Sentiment	4.72	n.a.	4.66 ($p < 0.001$)
Cosine Similarity	0.4287	0.2993 ($p < 0.001$)	0.3708 ($p < 0.001$)

Note. Review length is measured by the number of characters or words in the review text; sentiment is scored using OpenAI GPT-4.1; and cosine similarity is the average within-product pairwise cosine similarity between review embeddings. Because all fake reviews are 5-star reviews, sentiment is compared only between fake and authentic 5-star reviews. Each reported p -value is obtained from a Welch’s t -test comparing the mean for fake reviews with that for the corresponding authentic-review group.

Table 2 reports the review-level summary statistics of Data Source 1. We measure review length by counting the number of characters and words in each review. To measure review sentiment, we use OpenAI GPT-4.1 to assign each review a score between 1 and 5.⁴ To measure textual similarity,

⁴The prompt we use is as follows. System prompt: “You are an AI assistant that scores the sentiment of reviews on Amazon. Respond succinctly in the requested format.” User prompt: “Analyze the sentiment of reviews. What is the quality perceived by consumers? Please output a float number: X. X should be between 1 and 5. Review Text: {copy and paste the review text here}” We set the temperature to zero to increase replicability.

we embed each review using the all-MiniLM-L6-v2 model. For each product, we divide reviews into authentic and fake groups, calculate the pairwise cosine similarity across all unique pairs within each group, and compute the group-level mean cosine similarity. On average, fake reviews are significantly longer, more positive, and more textually similar than authentic reviews. The patterns observed in our sample are consistent with the prior findings discussed in Section 3.2.

Data Source 2 consists of product-level data collected from Amazon on September 29, 2025. To construct the sample, we compile a list of 159 popular search terms based on two reports from commercial analytics firms: (i) Exploding Topics (Cardillo, 2025) and (ii) Glimpse (Glimpse, 2025). For each search term, we record the products appearing on the first seven pages of Amazon’s search results, yielding a sample of 16,851 products.⁵ For each product, we collect the following information from Amazon’s interface: (1) AI summary paragraphs, (2) keywords assigned to the extracted topics and their corresponding sentiment labels, (3) the number of reviews that mention each topic, (4) the total number of reviews, (5) the rating distribution, (6) the average rating, (7) the product name, (8) the product price,⁶ (9) the product category,⁷ and (10) whether the product is sold by Amazon.⁸ Panel (a) of Figure 1 shows where variables (1)–(6) appear on Amazon’s interface, while Panel (c) shows the locations of variables (7)–(10). Among the 16,851 products, 14,479 have AI summaries. We focus on the sample with AI summaries in this paper, because our research questions (including whether AI summaries overrepresent fake reviews and whether they are systematically more positive for review manipulators) are not applicable to products without AI summaries. The middle panel of Table 1 reports summary statistics of Data Source 2. The total number of reviews in this source is much higher than that in Data Source 1. Both the number of reviews and product prices are highly right-skewed: their means substantially exceed their medians. Average ratings, by contrast, exhibit limited dispersion, concentrating between 4.3 (25th percentile) and 4.6 (75th percentile). Approximately 30% of the products in the sample are sold by Amazon.

⁵We stopped collecting products for a search term after recording more than 110 products, even if we had not yet reached the seventh page of search results.

⁶We download the price of the default size and color. We use the post-discount price throughout the paper.

⁷Amazon’s interface displays a multi-level category. When controlling for categories with fixed effects, we use the finest level of categories, so that we control for enough unobservable confounders. When conducting exact matching on categories, we match on the first-level category, so that we do not have to drop too many observations.

⁸The products without AI summaries typically have very few reviews. According to Amazon, an AI summary of reviews does not appear unless customer reviews contain enough shared mentions of common features or topics to support reliable synthesis (Levine, 2024).

Data Source 3 consists of review-credibility assessments obtained from RateBud. RateBud provides the review-credibility grades for 14,321 out of the 14,479 products with AI summaries in Data Source 2. Its underlying AI model evaluates review authenticity by analyzing features such as reviewer profile history, linguistic patterns, and posting timestamps. Table 3 reports the distribution of review-credibility grades. We find that most products receive grades A or B. Consistent with He et al. (2022b), the distribution of review-credibility grades for Amazon-sold products is concentrated in grade A, while the distribution of review-credibility grades for non-Amazon-sold products is more dispersed. Additionally, very few products with grades D or F are sold by Amazon. We use Amazon-sold status and RateBud review-credibility grades as complementary proxies for products’ exposure to fake reviews.

Table 3: Summary Statistics for Data Source 3 (RateBud)

Grade	Meaning	All		Amazon-Sold		Non-Amazon-Sold	
		Count	Percentage	Count	Percentage	Count	Percentage
A	Highly trustworthy	9,781	68.30%	3,591	77.90%	6,174	63.74%
B	Generally reliable	3,886	27.13%	901	19.54%	2,976	30.72%
C	Mixed signals	590	4.12%	113	2.45%	477	4.92%
D	Significant concerns	49	0.34%	5	0.11%	44	0.45%
F	High risk of manipulation	15	0.10%	0	0	15	0.15%
Total		14,321	100%	4,610	100%	9,686	100%

Note. RateBud assigns review-credibility grades A, B, C, D, or F, with lower grades indicating lower credibility. Percentages are calculated within columns. The *All* column includes the 14,321 products with available RateBud grades among the 14,479 products with AI summaries in Data Source 2. The *Amazon-Sold* and *Non-Amazon-Sold* columns sum to 14,296, rather than 14,321, because Amazon-sold status is missing for 25 out-of-stock products; these products remain in the *All* column.

Data Source 4 consists of historical sales rank data obtained from Keepa. Amazon assigns each product a sales rank within its first-level category, as illustrated in Panel (c) of Figure 1. We queried Keepa for the products in Data Source 2 and were able to retrieve sales rank history from February 17, 2022, through December 31, 2024, for 10,743 out of the 14,479 products with AI summaries. We retain products with observations both before and after the introduction of AI summaries and end up with 6,823 products in total. Because Keepa may record multiple sales-rank observations for the same product within a day, we construct a product-day panel by taking the arithmetic mean of all available sales-rank observations for each product on each calendar day. This panel is unbalanced: not all products have observations every day, and 14.2% of the observations are missing. Table 1 reports the summary statistics of sales ranks using product-day observations.

In Appendix E, we report their product-level summary statistics. Table 1 shows that sales ranks are highly right-skewed, consistent with the well-documented long-tail pattern in online markets (Brynjolfsson et al., 2011). Hence, in our analysis, we follow Chevalier and Mayzlin (2006) and use the logarithmic transformation of sales ranks as the dependent variable.

4 Theoretical Framework and Hypothesis Development

In this section, we first formally define the overrepresentation of fake reviews, and then develop testable hypotheses.

4.1 Definition of Overrepresentation

Our definition of overrepresentation of fake reviews follows the framework established in the political economy literature on group representation (Adukia et al., 2023; Anwar et al., 2022; Gomes and Kuehn, 2025). In this literature, a group is considered overrepresented when its representation in a given institutional or social context (e.g., children’s books, juries, or public-sector employment) exceeds its proportion in the reference population. We apply the same framework to AI review summaries, treating the underlying review corpus as the relevant population and the extent to which each review is reflected in the AI summary as its representation.

Formally, let j index products and i index reviews. Let F_{ij} be an indicator for whether review i of product j is fake. Let $\mathcal{K}_j$ be the set of common topics extracted by the AI summarization algorithm for product j . Define W_{ij} as the number of extracted topics mentioned by review i of product j :

$$W_{ij} \equiv \sum_{k \in \mathcal{K}_j} 1\{\text{review } i \text{ mentions topic } k\}.$$

Definition 1 (Overrepresentation of Fake Reviews). *AI summaries overrepresent fake reviews if, on average, fake reviews mention more extracted AI-summary topics than authentic reviews:*

$$\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0].$$

Intuitively, the AI summarization process can be viewed as allowing each review to cast one “vote” for every extracted topic that it mentions. Hence, a review that mentions more extracted topics receives greater representation by contributing to the sentiments of a larger number of topics. Accordingly, fake reviews are overrepresented when they receive more such representation per review, on average, than authentic reviews.⁹

4.2 Hypothesis Development

As explained in Section 3.1, AI summarization algorithms aim to extract the topics that appear most frequently across consumer reviews (Schermerhorn, 2023). Consistent with prior evidence that fake reviews are more mutually similar (Jindal and Liu, 2008; Mukherjee et al., 2012; Rayana and Akoglu, 2015; Wang et al., 2023; Gupta et al., 2024) and less concrete (Ott et al., 2011), Table 2 documents significantly higher within-product textual similarity among fake reviews in our sample. These patterns suggest that fake reviews tend to draw on a shared pool of generic product merits rather than on idiosyncratic details from personal experience. The presence of fake reviews inflates the prevalence of these generic topics, making them more likely to be selected by topic-extraction algorithms and featured in AI summaries. We refer to this mechanism as the *common-topic alignment channel*. A complementary *review-length channel* may reinforce this alignment. According to He et al. (2022b) and our summary statistics in Table 2, fake reviews on Amazon are longer, giving each review greater scope to mention more of these shared merits and thus match more topics in $\mathcal{K}_j$.¹⁰ Together, these two channels lead us to expect that an average fake review will mention more extracted topics (i.e., yield a higher W_{ij}) than an authentic review. We therefore hypothesize that AI summaries overrepresent fake reviews.

Hypothesis 1 (Overrepresentation of Fake Reviews). *On average, fake reviews mention more extracted AI-summary topics than authentic reviews, i.e., $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$.*

To test Hypothesis 1, we must identify fake reviews and determine which extracted topics a review mentions. For this purpose, we use the dataset compiled by Hollenbeck (Data Source 1).

⁹In political economy, the ideal situation is $\mathbb{E}[W_{ij} \mid F_{ij} = 1] = \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ (i.e., no overrepresentation of any group). But in our setting, it is socially beneficial to have $\mathbb{E}[W_{ij} \mid F_{ij} = 1] < \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ (i.e., the underrepresentation of fake reviews). In fact, the ideal situation is arguably $\mathbb{E}[W_{ij} \mid F_{ij} = 1] = 0$ (i.e., zero representation of fake reviews).

¹⁰Appendix F provides supporting evidence for both channels.

A major advantage of this dataset is that prior research has already pre-classified fake reviews, allowing us to test Hypothesis 1 directly.

However, Hollenbeck’s dataset is limited to a relatively small subset of products and reviews. To examine whether the same pattern extends to a broader set of products, we develop a complementary product-level hypothesis that can be tested using publicly observable data, eliminating the need to identify fake reviews. Let $f_j \equiv \mathbb{E}[F_{ij} \mid j]$ denote the share of fake reviews for product j , and let $w_j \equiv \mathbb{E}[W_{ij} \mid j]$ denote the average number of extracted topics mentioned per review for product j . Hypothesis 1 specifies a review-level condition equivalent to $\text{Cov}(W_{ij}, F_{ij}) > 0$. Aggregating W_{ij} and F_{ij} into product-level averages yields its product-level analogue, predicting that $\text{Cov}(w_j, f_j) > 0$. Hypothesis 2 then posits that this positive product-level correlation persists after controlling for product-level covariates.

Hypothesis 2 (Product-Level Overrepresentation of Fake Reviews). *Products with a higher share of fake reviews have a higher expected number of extracted topics mentioned per review, i.e., w_j increases with f_j , holding other product characteristics fixed.*

We note that Hypotheses 1 and 2 are non-nested. We therefore regard these two hypotheses as complementary.¹¹

Our next hypothesis concerns how the overrepresentation of fake reviews distorts product quality information. By definition, the overrepresentation of fake reviews shifts the aggregated sentiment for each topic toward positive fake reviews, thereby introducing an upward bias in the sentiment. This will be reflected in more positive sentiment for each topic. Because the AI summary paragraph is generated based on the sentiment of the extracted topics, such topic sentiment bias would also lead to a more favorable AI summary paragraph. This prediction motivates Hypothesis 3.

Hypothesis 3 (Distortion in Product Quality Information). *After controlling for average ratings, products with a higher share of fake reviews exhibit more positive sentiment across extracted topics and yield more favorable AI-generated summary paragraphs.*

¹¹Appendix G formally derives the relationship between Hypotheses 1 and 2, including the conditions under which they are equivalent.

Researchers have long recognized that online reviews can influence sales (Chevalier and Mayzlin, 2006; Zhu and Zhang, 2010; Sun, 2012; Lu et al., 2022). Because AI review summaries are displayed prominently above individual reviews and condense information from the review corpus, they likely also shape consumers’ perceptions and purchase decisions. If the overrepresentation of fake reviews makes AI summaries more favorable for products with greater exposure to review manipulation, the introduction of these summaries may shift demand toward such products relative to their competitors with less exposure. Hypothesis 4 formalizes this prediction. It characterizes a market distortion in which deceptive sellers achieve better sales outcomes at the expense of honest sellers. Such a market distortion may lead consumers to purchase suboptimal products.

Hypothesis 4 (Distortion in Market Outcomes). *Following the introduction of AI summaries of reviews, products with a higher share of fake reviews experience a relatively greater sales improvement than their competitors.*

5 Results

5.1 Testing H1: Review-Level Evidence for Overrepresentation of Fake Reviews

5.1.1 Results for Testing H1

The overrepresentation of fake reviews, as defined in Definition 1 (i.e., $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$), involves both F_{ij} (the indicator for fake reviews) and W_{ij} (the number of extracted topics mentioned by review i). There are two challenges: (i) fake reviews are hard to identify, and (ii) we cannot directly observe which topics each review mentions because the displayed keywords are merely topic labels. To address the first challenge, we use Data Source 1 and classify reviews as fake based on the three review-level criteria described in Section 3.3. To address the second challenge, we apply modern NLP methods to construct a proxy for W_{ij} .

Proxy for W_{ij} . We download from Amazon’s interface the keywords associated with each extracted AI-summary topic. We then determine whether a review mentions a given extracted topic using two complementary criteria. First, we perform case-insensitive, whole-phrase lexical matching between the review text and the keyword of each topic. Second, we apply sentence-level

semantic matching: we split each review into individual sentences and project both keywords and review sentences into embedding space using the all-MiniLM-L6-v2 model. This second criterion is satisfied if the cosine similarity between a keyword’s embedding and at least one sentence embedding in the review exceeds 0.8. A topic is defined as mentioned by a review if either the lexical or semantic criterion is met. Finally, we count the number of distinct extracted topics mentioned by review i of product j as a proxy for W_{ij} .

Table 4 reports the test results for Hypothesis 1. In Appendix A.4, we test Hypothesis 1 using regression specifications that include more flexible controls and product fixed effects. On average, fake reviews mention 0.40 topics, compared with 0.23 among all authentic reviews and 0.33 among authentic five-star reviews. This finding provides direct evidence that $\mathbb{E}[W_{ij} \mid F_{ij} = 1] > \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ —that is, AI summaries overrepresent fake reviews. The comparison with authentic five-star reviews indicates that the result is not solely driven by the fact that fake reviews are positive.

Table 4: Review-Level Analysis of Overrepresentation of Fake Reviews from Data Source 1

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Amazon-Extracted Topics (full sample)	0.4017	0.2335 ($p < 0.001$)	0.3336 ($p < 0.001$)
Amazon-Extracted Topics (excluding removed reviews)	0.4298	0.2380 ($p < 0.001$)	0.3384 ($p < 0.001$)
BERTopic-Extracted Topics	1.0082	0.7388 ($p < 0.001$)	0.8577 ($p < 0.001$)
Top2Vec-Extracted Topics	0.2595	0.1517 ($p < 0.001$)	0.2289 ($p < 0.001$)

Note. This table tests Hypothesis 1 by comparing mean W_{ij} for fake reviews with the corresponding means for all authentic reviews and authentic five-star reviews. Topics are obtained from Amazon’s interface or extracted using BERTopic or Top2Vec, and topic mentions are identified using lexical and semantic matching. We also try excluding reviews subsequently removed by Amazon as a robustness check. Each reported p -value is obtained from a Welch’s t -test comparing the mean for fake reviews with that for the corresponding authentic-review group.

5.1.2 Robustness Checks

We conduct several robustness checks, detailed below.

1. [Hollenbeck \(2026\)](#) compared his dataset, collected in 2019–2020, with the dataset from [Hou et al. \(2024\)](#), which includes reviews through September 2023, and found that some of the reviews in his dataset were removed by Amazon (around 13%).¹² Because Amazon’s AI summarization algorithm was launched in mid-2023, the removed reviews may not have been used in the topic-extraction step. We therefore repeat our analysis by excluding these removed reviews and report the results in the second row of Table 4. The findings remain

¹²Amazon monitors reviews over time and removes those it suspects to be fake.

robust, and the difference in W_{ij} between authentic and fake reviews is actually slightly larger in this robustness check.

2. Because Data Source 1 contains a small subset of reviews per product collected in 2019–2020, while the AI summary keywords were collected on April 23, 2026, one may be concerned that the keywords may not correspond well to the reviews in Data Source 1. To address this concern, we apply two topic extraction models, BERTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020), to the reviews in Data Source 1. We explain the details of these two models in Appendix A.1. The results using these two alternative models are consistent with those from Amazon-extracted topics.
3. When determining whether a review mentions an extracted topic:
 - (a) **Alternative Embedding Model:** In Appendix A.2, we rerun our analysis using paraphrase-mpnet-base-v2 to generate sentence and keyword embeddings. The results remain unchanged.
 - (b) **Alternative Similarity Thresholds:** While our main result relies on a cosine similarity threshold of 0.8, we evaluate alternative thresholds of 0.4 and 0.6 in Appendix A.3. The findings remain qualitatively robust.

5.2 Testing H2: Product-Level Evidence for Overrepresentation of Fake Reviews

The product-level test of Hypothesis 2 complements the review-level test of Hypothesis 1 in two ways. First, it extends our analysis to a larger, more representative sample. Second, it relies directly on the outputs of the AI summarization algorithms actually deployed by Amazon. Therefore, this analysis mitigates the concerns raised in Section 5.1.2.

To test Hypothesis 2, we make use of Data Sources 2 and 3. We compare the average number of extracted topics mentioned per review at the product level (w_j) across products with a higher versus lower share of fake reviews (f_j). We approximate $w_j \equiv \mathbb{E}[W_{ij} \mid j]$ using its sample average,

$\hat{w}_j$, as follows:

$$\begin{aligned}\hat{w}_j &\equiv \frac{1}{N_j} \sum_{i=1}^{N_j} W_{ij} \equiv \frac{1}{N_j} \sum_{i=1}^{N_j} \sum_{k \in \mathcal{K}_j} 1\{\text{review } i \text{ mentions topic } k\} \\ &= \frac{1}{N_j} \sum_{k \in \mathcal{K}_j} \sum_{i=1}^{N_j} 1\{\text{review } i \text{ mentions topic } k\}.\end{aligned}$$

For each product j , the Amazon user interface displays the number of reviews that mention each topic (i.e., $\sum_{i=1}^{N_j} 1\{\text{review } i \text{ mentions topic } k\}$), as well as the total number of reviews (i.e., N_j).

We use the following proxies to identify products with different shares of fake reviews (f_j).

1. **Amazon-sold products:** As Amazon rarely manipulates reviews (He et al., 2022b), first-party Amazon-sold items serve as a proxy for products with a *lower* f_j .
2. **Products with low RateBud review credibility:** We collect review-credibility grades from RateBud (Table 3), ranging from A (“Highly trustworthy”) to F (“High risk of manipulation”). Our baseline specifications regard grades C, D, and F as low review credibility (i.e., a *higher* f_j). Appendix B.1 shows that the results remain robust to alternative definitions of low review credibility: a stricter definition that includes only grades D and F, and a more inclusive definition that includes all non-A grades (grades B, C, D, and F).

Let j index products, c index product categories, k index search terms, and n denote the page number on which product j appears in the search results. We estimate the following product-level regression,

$$w_j = \beta I_j + \mathbf{X}_j' \gamma + \phi_c + \psi_{kn} + \epsilon_j, \quad (1)$$

where I_j is an indicator for products with higher (or lower) shares of fake reviews, and $\mathbf{X}_j$ is a vector of flexible control variables, including the number of extracted topics, average rating, review count, and an indicator for whether the product is shipped by Amazon. Additionally, we include category fixed effects, ϕ_c , to absorb unobserved differences across product categories, and search term $\times$ page number fixed effects, ψ_{kn} , to restrict comparisons to products appearing on the same page of search results for the same query. The coefficient of interest, β , therefore captures the difference in w_j between products with higher and lower shares of fake reviews. In Appendix B.4, we conduct

robustness checks that include more flexible controls for the rating distribution; our main findings on β remain unchanged.

5.2.1 Results for Testing H2

Table 5 presents the results for a series of specifications. We compare Amazon-sold products and non-Amazon-sold products in Column (1), and products with low-review-credibility grades on RateBud and other products in Column (2). In Column (3), we compare products with low-review-credibility grades on RateBud with the other products within the subsample of non-Amazon-sold products. Across these three specifications, products with higher f_j consistently exhibit higher w_j . This positive correlation between w_j and f_j at the product level supports Hypothesis 2.

Table 5: Product-Level Analysis of Overrepresentation of Fake Reviews

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Amazon-Sold	-0.126*** (0.00683)			-0.0611*** (0.0153)			-0.135*** (0.00778)		-0.131*** (0.00774)
Low Review Credibility		0.351*** (0.0152)	0.319*** (0.0179)		0.242*** (0.0204)	0.216*** (0.0231)		0.253*** (0.0137)	0.246*** (0.0135)
# Topics							0.0224*** (0.00213)	0.0268*** (0.00215)	0.0276*** (0.00213)
Avg Rating							0.170*** (0.0129)	0.172*** (0.0130)	0.178*** (0.0128)
# Reviews							-1.65e-06*** (1.21e-07)	-1.88e-06*** (1.24e-07)	-1.70e-06*** (1.23e-07)
Shipped from Amazon							0.112*** (0.0102)	0.0435*** (0.00948)	0.106*** (0.0101)
Constant	0.454*** (0.00387)	0.397*** (0.00326)	0.437*** (0.00423)	0.354*** (0.00744)	0.312*** (0.00662)	0.328*** (0.00785)	-0.539*** (0.0593)	-0.572*** (0.0597)	-0.618*** (0.0590)
< 8 Topics Only	No	No	No	Yes	Yes	Yes	No	No	No
Excluding Amazon-Sold	No	No	Yes	No	No	Yes	No	No	No
Search Term $\times$ Page No. FE	No	No	No	No	No	No	Yes	Yes	Yes
Category FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	14416	14284	9658	2537	2528	1916	13421	13270	13270
R^2	0.023	0.036	0.032	0.006	0.053	0.044	0.475	0.477	0.489

Note. The dependent variable is w_j , the average number of extracted topic mentions per review. *Amazon-Sold* is an indicator for products sold by Amazon and serves as a proxy for a lower share of fake reviews. *Low Review Credibility* is an indicator for products receiving a C, D, or F grade from RateBud and serves as a proxy for a higher share of fake reviews. Columns (4)–(6) restrict the sample to products with fewer than eight displayed topics; these products are therefore not constrained by Amazon’s display limit.

Standard errors are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

One concern is that there may be an upper limit on the number of topics that can be displayed on Amazon’s interface, which may provide an alternative explanation for our finding. To address this concern, Columns (4)–(6) of Table 5 restrict the sample to the subset of products for which the number of displayed topics falls below eight, which is the maximum number of topics observed

in the data. Therefore, the number of extracted common topics for products in this subsample is not constrained by the display limit. The results on *Amazon-Sold* and *Low Review Credibility* remain qualitatively unchanged. Columns (7)–(9) further include a set of controls and search term $\times$ page number and category fixed effects; the positive correlation between w_j and f_j remains robust.

Another concern is that one criterion RateBud uses to assess review credibility relies on AI-based comparisons of review text with patterns associated with known fake, suspicious, or AI-generated reviews. Because this criterion may itself capture textual repetition, it could mechanically generate a positive correlation with w_j . To address this concern, we reconstruct the overall credibility grades by using only criteria other than textual or linguistic features of reviews. The results, reported in Appendix B.2, remain unchanged.

A few remarks are in order regarding how the product-level analysis complements the review-level analysis. First, while the review-level test provides direct evidence of the overrepresentation of fake reviews, it is restricted to a relatively small sample of products and a subset of reviews per product. The product-level test extends our analysis to a much broader set of products.

Second, the review-level test relies on inferring topic mentions within each review via lexical and semantic matching. The product-level test bypasses this approximation because Amazon directly reports the number of reviews mentioning each displayed topic, enabling us to analyze the actual outputs of the AI summarization system. Consequently, while the two analyses navigate different trade-offs between internal validity and external relevance, both yield consistent conclusions.

5.3 Testing H3: Distortion in Product Quality Information

Hypothesis 3 states that, *compared with their competitors, products with a higher share of fake reviews have more positive sentiment labels for extracted topics and more positive sentiment in summary paragraphs*. Intuitively, this means that the sentiment of AI summaries is biased in favor of review manipulators. To test this hypothesis, we consider two product-level dependent variables: *topic sentiment* and *paragraph sentiment*.

Topic sentiment. The sentiment label corresponding to each extracted topic is directly observable on Amazon (the positive, negative, or neutral marker preceding each keyword in Panel (a) of Figure 1). For each topic (represented by a keyword), we code positive sentiment as 1, negative

sentiment as -1 , and neutral sentiment as 0 . Then, *topic sentiment* is the average of these values across topics within each product.

Paragraph sentiment. To measure the sentiment in summary paragraphs, we use OpenAI GPT-4.1. We prompt the model to rate, on a 1–5 scale aligned with Amazon star ratings, the product quality conveyed to consumers by each summary paragraph.¹³ In a robustness check, we apply bert-base-multilingual-uncased-sentiment, which is a specialized language model fine-tuned on the task of predicting review ratings; we report the findings in Appendix C.1 and our main results remain unchanged.

Recall that j indexes products, c indexes categories, k indexes search terms, and n denotes the search result page number on which product j appears. We estimate the following specification,

$$Sentiment_j = \beta I_j + \mathbf{X}_j' \gamma + \phi_c + \psi_{kn} + \epsilon_j, \quad (2)$$

where $Sentiment_j$ represents either topic sentiment or paragraph sentiment for product j , I_j is an indicator for products with higher (or lower) shares of fake reviews, and $\mathbf{X}_j$ is a vector of product and review characteristics, including average rating, review count, and an indicator for whether the product is shipped by Amazon. As in Equation (1), we include category fixed effects and search term $\times$ page number fixed effects. The parameter β captures the conditional difference in AI-summary sentiment between products with higher versus lower fake review shares. In Appendix C.5, we include more flexible controls for the rating distribution and our main findings remain robust.

5.3.1 Results for Testing H3

The results are shown in Columns (1)–(4) of Table 6. Columns (1) and (2) use Amazon-sold status as a proxy for a lower share of fake reviews, whereas Columns (3) and (4) use low-review-credibility grades on RateBud as a proxy for a higher share of fake reviews. The dependent variable is topic sentiment in Columns (1) and (3), and paragraph sentiment in Columns (2) and (4).

¹³The prompt we use is as follows. System prompt: “You are an AI assistant that scores the sentiment of AI summary of online reviews on Amazon. Respond succinctly in the requested format.” User prompt: “Analyze the sentiment of AI summary of reviews. What is the quality perceived by consumers? Please output a float number: X. X should be between 1 and 5. AI Summary: {The content of AI summary}” We set the temperature to zero to increase replicability.

Across all specifications, the results support Hypothesis 3. Amazon-sold products, which are products with a lower share of fake reviews, receive less positive topic and paragraph sentiment, while low-review-credibility products receive more positive topic and paragraph sentiment, conditional on the full set of controls and fixed effects. These findings show that greater exposure to fake reviews is associated with more positive sentiment in Amazon’s AI summaries conditional on average ratings.

Table 6: Distortion in Product Quality Information

	Main Specifications				Robustness Checks			
	(1) Topic	(2) Paragraph	(3) Topic	(4) Paragraph	(5) Topic	(6) Paragraph	(7) Topic	(8) Paragraph
Amazon-Sold	-0.0677*** (0.00510)	-0.111*** (0.00949)			-0.0670*** (0.00513)	-0.110*** (0.00953)		
Low Review Credibility			0.0428*** (0.00897)	0.113*** (0.0166)	0.0386*** (0.00891)	0.106*** (0.0166)	0.0308** (0.0102)	0.0915*** (0.0190)
Avg Rating	0.702*** (0.00845)	1.182*** (0.0157)	0.702*** (0.00856)	1.185*** (0.0159)	0.706*** (0.00850)	1.191*** (0.0158)	0.668*** (0.0105)	1.137*** (0.0195)
# Reviews	-2.49e-07** (7.92e-08)	-7.21e-07*** (1.47e-07)	-3.66e-07*** (8.13e-08)	-9.28e-07*** (1.51e-07)	-2.74e-07*** (8.10e-08)	-7.78e-07*** (1.51e-07)	-5.25e-07*** (1.27e-07)	-1.09e-06*** (2.36e-07)
Shipped from Amazon	0.0794*** (0.00665)	0.103*** (0.0124)	0.0467*** (0.00624)	0.0488*** (0.0116)	0.0788*** (0.00667)	0.101*** (0.0124)	0.0752*** (0.00750)	0.0995*** (0.0139)
Constant	-2.468*** (0.0381)	-1.119*** (0.0710)	-2.464*** (0.0386)	-1.126*** (0.0716)	-2.485*** (0.0383)	-1.161*** (0.0713)	-2.314*** (0.0470)	-0.921*** (0.0874)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13459	13459	13307	13307	13307	13307	9071	9071
R ²	0.560	0.532	0.556	0.531	0.563	0.536	0.573	0.546

Note. *Topic Sentiment* is constructed by coding each topic-level sentiment label as 1 for positive, 0 for neutral, and -1 for negative and then averaging across the extracted topics of each product. *Paragraph Sentiment* is the 1–5 score assigned by OpenAI GPT-4.1 to the product quality that consumers would perceive from the summary paragraph. *Amazon-Sold* is an indicator for products sold by Amazon and proxies for a lower share of fake reviews. *Low Review Credibility* is an indicator for products receiving a C, D, or F grade from RateBud and proxies for a higher share of fake reviews. Columns (7)–(8) exclude Amazon-sold products.

Standard errors are in parentheses; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Columns (5)–(8) of Table 6 present two robustness checks. Columns (5) (topic sentiment) and (6) (paragraph sentiment) include indicators for both low-review-credibility products and Amazon-sold products. Columns (7) (topic sentiment) and (8) (paragraph sentiment) replicate the stricter comparison in Columns (3) and (6) of Table 5 by restricting the sample to non-Amazon-sold products. Both robustness checks yield consistent results.

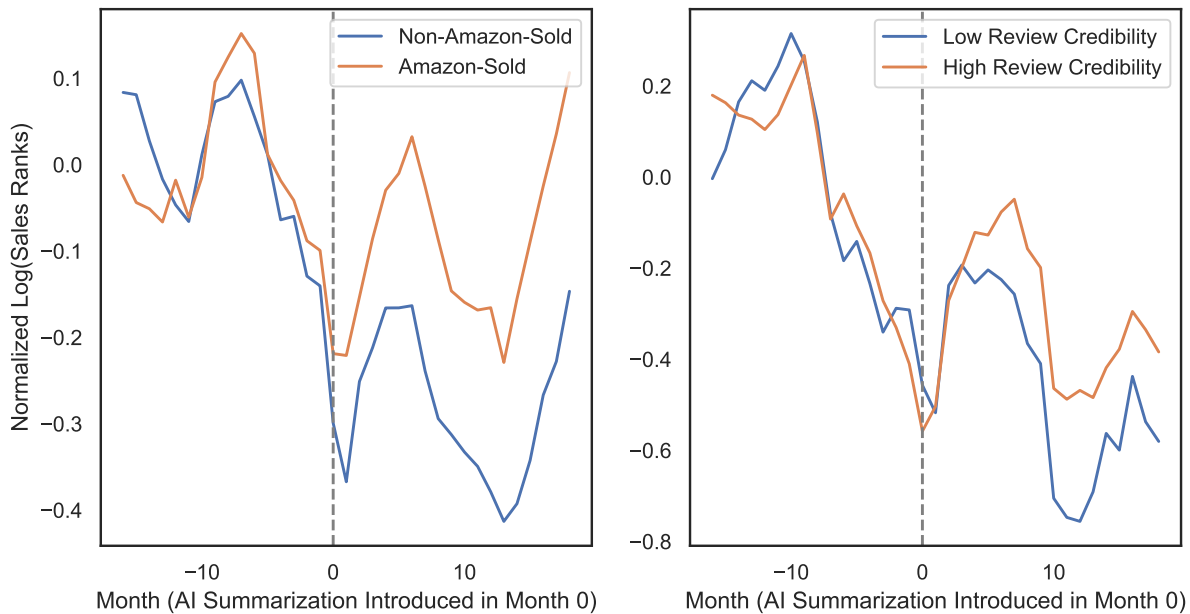
5.4 Testing H4: Distortion in Market Outcomes

Hypothesis 4 posits that review manipulators experience an increase in sales relative to their competitors following the introduction of AI summaries. Ideally, testing this hypothesis requires

sales data. However, because sales data are proprietary, we follow prior literature and use Amazon’s sales rank as a proxy for sales volume. We set June 12, 2023, as the AI summary introduction date based on the earliest published media report (Palmer, 2023). In Appendix D.2, we show that our results are robust to alternative event dates, including Amazon’s official announcement on August 14, 2023.

To ensure comparability across product groups with different shares of fake reviews, we perform exact matching on categories and the interaction of search terms and page numbers. Specifically, we define the matching strata as category $\times$ search term $\times$ page. We retain strata containing products from both groups, ensuring that comparisons are strictly made among products that belong to the same category and appear on the same search result page (i.e., sharing the same search term and page number). Within each retained stratum, products in one group receive unit weights, while products in the comparison group are reweighted so that the sum of weights is equal across the two groups within each stratum. All subsequent analyses incorporate these matching weights.

Figure 2: Descriptive Evidence for the Impact on Sales Ranks



Note. The y-axis reports log sales rank, normalized by subtracting its mean during the pre-AI-summary period. To ensure comparability between the two subgroups, we exact-match products on category and search term $\times$ page number. Non-Amazon-sold products and products with low review credibility proxy for higher shares of fake reviews.

We start by presenting descriptive evidence in Figure 2. The left sub-figure compares Amazon-sold versus non-Amazon-sold products, whereas the right sub-figure compares the high- versus low-review-credibility (from RateBud) products. In both sub-figures, normalized log sales ranks

evolve in parallel for the two groups before the introduction of AI summaries but diverge thereafter. More specifically, products with higher shares of fake reviews experience a decline in log sales ranks relative to their competitors. Because a lower sales rank indicates higher sales, this divergence is consistent with review manipulators gaining sales after the introduction of AI summaries.

5.4.1 Results for Testing H4

To formally test Hypothesis 4, we estimate the following specification at the product-day level. Let j index products, t index calendar dates, and m index relative months from the introduction of AI summaries. Following Chevalier and Mayzlin (2006), we use $\log(\text{Sales_Rank}_{jt})$ as our dependent variable.

$$\log(\text{Sales_Rank}_{jt}) = \sum_m \beta_m (I_j \times \phi_{mt}) + \alpha_j + \omega_t + \epsilon_{jt}, \quad (3)$$

where I_j is an indicator for whether product j belongs to a subgroup with a higher (or lower) share of fake reviews, as defined in Section 5.2; ϕ_{mt} indicates whether date t falls in relative month m ; α_j and ω_t denote product and date fixed effects, respectively. The parameter of interest is β_m . It is worth highlighting that Equation (3) should not be viewed as a standard DiD regression because it violates the Stable Unit Treatment Value Assumption (SUTVA). Sales rank is inherently relative: sales gains by products with higher fake-review exposure may both divert consumer demand from lower fake-review products and mechanically depress their ranks even if their sales remain constant. Hence, β_m should be interpreted strictly as measuring the relative competitive divergence between products with higher and lower shares of fake reviews.¹⁴

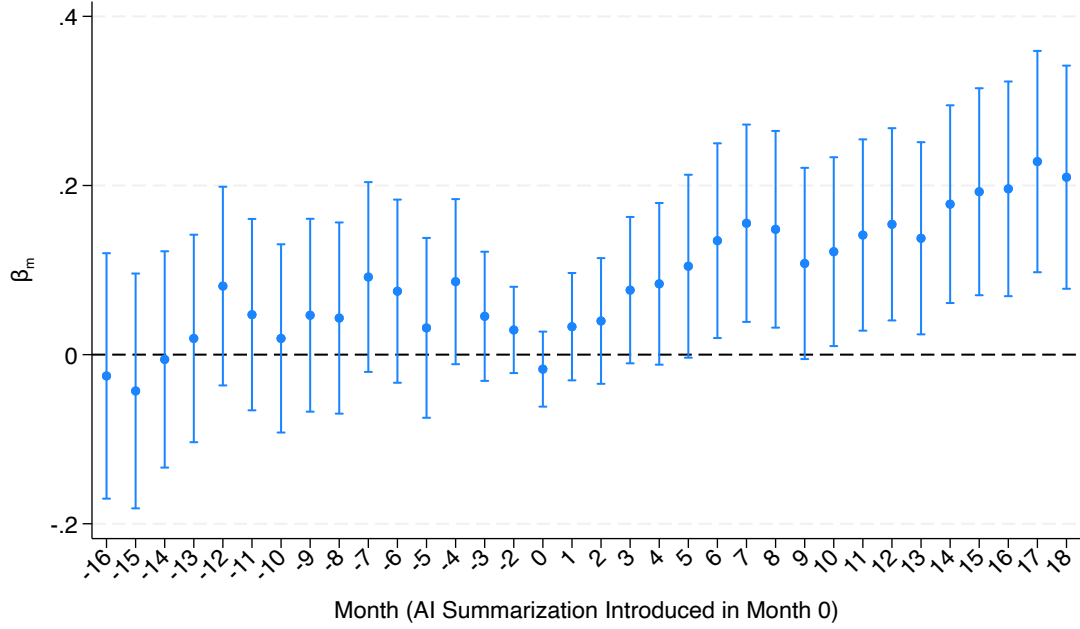
A positive β_m for $m > 0$ provides support for Hypothesis 4, as it indicates that products with greater exposure to fake reviews have better relative sales performance after the introduction of the AI summary. We cluster standard errors at the product level to account for potential serial correlation within products. Due to space constraints, the estimates are reported in Appendix D.1. As a robustness check, we also estimate the same model on the full sample that includes products without AI summaries as well; the findings remain unchanged and are reported in Appendix D.3.

We plot the estimated β_m coefficients for the regression using the Amazon-sold indicator in Figure 3. Prior to the introduction of AI summaries, the estimated pre-treatment coefficients

¹⁴Visually, β_m measures the difference between the two lines in Figure 2 in month m with flexible fixed effects and higher statistical power.

are statistically indistinguishable from zero and exhibit no systematic pattern, supporting the parallel-trends assumption. Following the introduction of AI summaries, the relative sales rank of Amazon-sold products increases over time (indicating lower relative sales volume), providing support for Hypothesis 4.

Figure 3: Comparison between Amazon-Sold and Non-Amazon-Sold Products

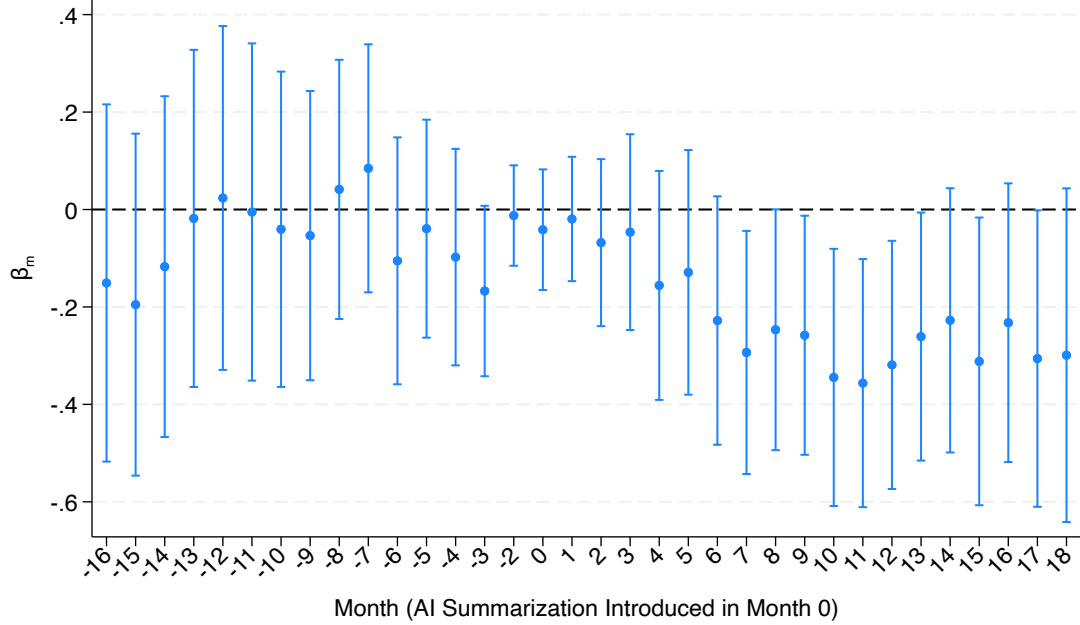


Note. The figure plots the estimates of β_m and their 95% confidence intervals from the regression in Equation (3). The indicator I_j equals one if product j is sold by Amazon and thus proxies for lower shares of fake reviews. The x -axis reports m , with Month 0 denoting the introduction month and Month -1 omitted as the reference period. Products are matched exactly on category and search term $\times$ page number, and standard errors are clustered at the product level.

We plot the corresponding β_m estimates for the regression using the low-review-credibility indicator in Figure 4. Analogous to Figure 3, the pre-treatment coefficients display no differential trend between high- and low-review-credibility products. Following the introduction, the relative sales rank of low-review-credibility products decreases over time (reflecting higher relative sales volume), offering further empirical support for Hypothesis 4.

Overall, these findings not only support Hypothesis 4, but also indicate that the overrepresentation of fake reviews could generate persistent downstream market distortions, after we control for the pre-existing differences before the introduction of AI summaries, including those caused by fake reviews but not through AI summaries.

Figure 4: Comparison between Low-Review-Credibility Products and Other Products



Note. The figure plots the estimates of β_m and their 95% confidence intervals from the regression in Equation (3). The indicator I_j equals one if product j receives a C, D, or F grade from RateBud and thus proxies for higher shares of fake reviews. The x -axis reports m , with Month 0 denoting the introduction month and Month -1 omitted as the reference period. Products are matched exactly on category and search term $\times$ page number, and standard errors are clustered at the product level.

Note that instead of exhibiting an immediate step-change at Month 0, these two figures show that the divergence in relative sales ranks is gradual and widens progressively over the 18-month post-treatment window. This dynamic pattern could be due to the gradual rollout of AI summarization to an increasing number of products. It may also reflect consumers' increasing reliance on AI summaries and sellers' increased investment in fake reviews as such summaries raise the returns to manipulation. These channels are not mutually exclusive, and distinguishing between them is not our objective; our analysis tests whether AI summaries distort market outcomes in favor of review manipulators.

6 Conclusion

This paper documents an unintended consequence of AI-generated review summaries: they systematically overrepresent fake reviews. This is due to the unintended alignment between the objective of AI summarization algorithms and the textual features of fake reviews. AI summarization algorithms are designed to summarize common topics among user-generated reviews, while

fake reviews are disproportionately concentrated on common topics. In the empirical exercise, we demonstrate that AI summaries overrepresent fake reviews through complementary review-level and product-level analyses. We then empirically test two effects of the overrepresentation. First, we show that products with a higher share of fake reviews receive more positive AI summaries, even after controlling for average ratings. Second, we find that after the introduction of AI summaries, these products with higher shares of fake reviews experience higher sales than their competitors.

These findings have important implications for platform governance. Platforms deploy AI summaries to reduce information overload and enhance the consumer experience; however, when AI summaries render manipulated products more attractive, short-run conversion gains may come at the expense of long-run consumer trust and platform integrity. The main challenge is therefore not merely how to summarize text accurately, but how to do so robustly in adversarial environments.

These results offer a broader, cautionary lesson regarding the design and deployment of AI in information environments contaminated by strategic manipulation. In such environments, a summarization algorithm that accurately identifies common topics can unintentionally amplify deceptive content that platforms ought to discount. This bias does not require training on protected characteristics, nor does it require inaccurate predictions for any group. It arises from a mismatch between the algorithm's objective—identifying common topics—and consumers' objective—getting reliable product information. Simply improving the accuracy of the current algorithms could make the bias even worse. We hope this research will inspire researchers across disciplines to work toward innovative algorithms that are robust to manipulated information.

Our market distortion finding also warrants broader societal attention. Prior research shows that AI summaries can increase purchase rates ([Wang et al., 2025](#)), giving platforms an economic incentive to adopt them. This may help explain the growing use of AI summarization. Our findings suggest a possible countervailing effect: AI summaries systematically overrepresent fake reviews and may consequently steer consumers toward suboptimal products. It is plausible that this distortion could harm consumers and, if so, warrant regulatory intervention. Determining whether, and to what extent, this distortion harms consumers is therefore an important question for future research.

While our empirical analysis focuses on Amazon, the underlying mechanism should extend to other settings where summarization systems treat frequency or recurrence as a proxy for relevance, such as AI review summaries on the Apple App Store. Assessing generalizability across other platforms remains an important avenue for future research.

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Appendix A Robustness Checks for Testing H1: Review-Level Evidence for Overrepresentation of Fake Reviews

A.1 Details on Other Topic Models

We also use BERTopic (Grootendorst, 2022) and Top2Vec (Angelov, 2020) to extract topics from the reviews in our dataset. Both models use UMAP to reduce the dimensionality of the embeddings (generated by all-MiniLM-L6-v2) and HDBSCAN to cluster the embeddings. Then, BERTopic uses TF-IDF to identify salient n-grams as the keyword assigned to each cluster of embeddings, while Top2Vec constructs a topic vector as the centroid of the review embeddings for each cluster, and then ranks candidate words and phrases by their cosine similarity to this topic vector. In each method, we retain the top representative phrase for each topic after filtering out generic marketplace terms and near-duplicate expressions, yielding a product-specific keyword set.

A.2 Alternative Embedding Models

We rerun the analyses with a larger embedding model named paraphrase-mpnet-base-v2. The results shown in Table A.1 are consistent with those reported in Table 4 in the main text.

Table A.1: Review-Level Analysis with paraphrase-mpnet-base-v2

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Amazon-Extracted Topics (full sample)	0.4046	0.2347 ($p < 0.001$)	0.3353 ($p < 0.001$)
Amazon-Extracted Topics (excluding removed reviews)	0.4325	0.2402 ($p < 0.001$)	0.3388 ($p < 0.001$)

A.3 Alternative Thresholds

When determining how many extracted topics each review mentions, we use semantic matching that treats a keyword as mentioned if the cosine similarity between the keyword embedding and at least one sentence embedding in the review exceeds a threshold. In the main text, we set the threshold at 0.8. Here, we report the results using thresholds of 0.6 and 0.4. The results with the thresholds of 0.6 and 0.4 are reported in Table A.2. They are consistent with the results in Table 4 in the main text using a threshold of 0.8.

Table A.2: Review-Level Analysis with the Thresholds of 0.6 and 0.4 in Semantic Matching

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
Amazon-Extracted Topics (0.6, full sample)	0.5083	0.2947 ($p < 0.001$)	0.4219 ($p < 0.001$)
Amazon-Extracted Topics (0.6, excluding removed reviews)	0.5342	0.2997 ($p < 0.001$)	0.4240 ($p < 0.001$)
Amazon-Extracted Topics (0.4, full sample)	1.2573	0.7525 ($p < 0.001$)	1.0514 ($p < 0.001$)
Amazon-Extracted Topics (0.4, excluding removed reviews)	1.2892	0.7591 ($p < 0.001$)	1.0615 ($p < 0.001$)

A.4 Regression Specifications

We complement the difference-in-means comparisons in Table 4 with review-level regressions of W_{ij} on F_{ij} . Specifically, we estimate the following specification:

$$W_{ij} = \beta F_{ij} + \gamma R_{ij} + \alpha_j + \epsilon_{ij},$$

where F_{ij} is an indicator for whether review i of product j is fake, R_{ij} is the review's star rating, and α_j denotes product fixed effects. The coefficient of interest, β , measures the difference between fake and authentic reviews in the number of Amazon-extracted topics mentioned per review. When product fixed effects are included, β is identified by comparing fake and authentic reviews of the same product. The fixed effects also absorb product-level factors common to all reviews of a product, including differences in the set of topics displayed by Amazon.

Table A.3 reports the results. Column (1) includes only the fake-review indicator and reproduces, up to rounding, the unconditional difference reported in Table 4: fake reviews mention 0.168 more extracted topics per review than authentic reviews. Column (2) adds product fixed effects, and Column (3) further clusters standard errors at the product level. Column (4) controls for review rating, while Column (5) restricts the sample to five-star reviews. The coefficient on the fake-review indicator remains positive and statistically significant across all specifications. These results support Hypothesis 1 and indicate that the observed difference is not solely attributable to cross-product heterogeneity or differences in review ratings.

Table A.3: Review-Level Regression Analysis of Overrepresentation of Fake Reviews

	(1) W_{ij}	(2) W_{ij}	(3) W_{ij}	(4) W_{ij}	(5) W_{ij}
F_{ij}	0.168*** (0.00468)	0.166*** (0.00506)	0.166*** (0.0144)	0.0854*** (0.0118)	0.110*** (0.0161)
Review Rating				0.0317*** (0.00403)	
Constant	0.234*** (0.00240)	0.234*** (0.00233)	0.234*** (0.00378)	0.155*** (0.0127)	0.296*** (0.0145)
Product FE	No	Yes	Yes	Yes	Yes
SE Clustered at Product Level	No	No	Yes	Yes	Yes
Five-Star Reviews Only	No	No	No	No	Yes
Observations	80120	80120	80120	80120	23463
R^2	0.016	0.154	0.154	0.158	0.220

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix B Robustness Checks for Testing H2: Product-Level Evidence for Overrepresentation of Fake Reviews

B.1 Alternative Cutoffs

RateBud grades are ordered from A, indicating the highest review credibility, to F, indicating the lowest. Our baseline analysis defines products receiving grades C, D, or F as having low review credibility. Tables A.4 and A.5 re-estimate the product-level analysis using, respectively, a stricter definition that includes only grades D and F and a more inclusive definition that includes grades B, C, D, and F.

Under both definitions, the coefficient on *Low Review Credibility* remains positive and statistically significant at the 0.1% level across all specifications. Together with the baseline definition, these analyses examine three of the four possible definitions. We do not examine an F-only definition because few products receive an F grade. The consistency of the results across these definitions suggests that our findings are not driven by the choice of cutoff.

B.2 Excluding Review Content Quality from the RateBud Measure

A potential concern with the RateBud-based analysis is that its Review Content Quality criterion may partly reflect similarities in the wording, structure, or content of reviews for the same product. Therefore, a review-credibility measure that incorporates within-product textual similarity could be mechanically associated with $\hat{w}_j$.

To address this concern, we exclude Review Content Quality and construct two alternative credibility scores as weighted averages of the remaining RateBud component scores: one using RateBud’s original component weights and the other assigning equal weight to each component. For both measures, higher values indicate greater review credibility. Under each measure, we classify a product as having low review credibility when its score is below the 25th percentile of the corresponding score distribution. By construction, neither measure depends on whether reviews for the same product resemble one another in their wording, structure, or content patterns. The remaining criteria do not create the same mechanical relationship with $\hat{w}_j$. This targeted exclusion therefore addresses the stated measurement concern while retaining the other dimensions of review credibility, avoiding an unnecessary loss of measurement precision and statistical power.

Tables A.6 and A.7 report the results using RateBud’s original weights and equal weights, respectively. The coefficient on *Low Review Credibility* remains positive and statistically significant at the 0.1% level in all six specifications. The results continue to hold when we exclude Amazon-sold products, restrict the sample to products with fewer than eight displayed topics, include the full set of controls and fixed effects, and control jointly for Amazon-sold status. Thus, the product-level association between exposure to review manipulation and $\hat{w}_j$ is not driven by RateBud’s *Review Content Quality* criterion or by the within-product textual similarity that this criterion may capture.

B.3 Share of Time in the Buy Box

One concern is that products sold by Amazon when we collected the data may not always be sold by Amazon. On Amazon, different sellers of a product share the same product page and compete to be the featured offer in the Buy Box.¹⁵ To deal with this concern, we collect the history of the featured offer in the Buy Box from Keepa. We confirm that Amazon occupied the Buy Box for Amazon-sold products during 80% of the observed period, compared with 6% for non-Amazon-sold products. This history is only available for a subset of products, so we still use Amazon-sold products as the proxy in the main text.

We use the share of time during which Amazon occupies each product's Buy Box as a proxy for lower exposure to fake reviews. We replace the Amazon-sold indicator with the continuous Amazon share variable. Table A.8 presents the result, which is largely consistent with that shown in the main text.

B.4 Alternative Controls

We include an alternative set of controls to check the robustness of the findings in the main text. We control for additional characteristics of the rating distribution. The results reported in Table A.9 remain consistent with the findings in the main text.

B.5 Excluding Products with Low Average Ratings

The average ratings of most products are above four. One concern is that the results may be driven by products with low average ratings. Therefore, we exclude products with average ratings below four. The results reported in Table A.10 are consistent with the findings in the main text.

¹⁵<https://sell.amazon.com/blog/what-is-an-asin>

B.6 Figures and Tables

Table A.4: Product-Level Analysis Using Grades D/F to Define Low Review Credibility

	(1)	(2)	(3)	(4)	(5)	(6)
Low Review Credibility	0.485*** (0.0484)	0.434*** (0.0534)	0.529*** (0.0685)	0.478*** (0.0735)	0.337*** (0.0439)	0.310*** (0.0435)
Amazon-Sold						-0.134*** (0.00783)
# Topics					0.0224*** (0.00216)	0.0232*** (0.00214)
Avg Rating					0.162*** (0.0131)	0.168*** (0.0130)
# Reviews					-1.90e-06*** (1.25e-07)	-1.72e-06*** (1.24e-07)
Shipped from Amazon					0.0471*** (0.00959)	0.111*** (0.0102)
Constant	0.411*** (0.00324)	0.452*** (0.00417)	0.333*** (0.00639)	0.348*** (0.00751)	-0.489*** (0.0602)	-0.537*** (0.0595)
< 8 Topics Only	No	No	Yes	Yes	No	No
Excluding Amazon-Sold	No	Yes	No	Yes	No	No
Search Term $\times$ Page No. FE	No	No	No	No	Yes	Yes
Category FE	No	No	No	No	Yes	Yes
Observations	14284	9658	2528	1916	13270	13270
R^2	0.007	0.007	0.023	0.022	0.465	0.477

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.5: Product-Level Analysis Using Grades B/C/D/F to Define Low Review Credibility

	(1)	(2)	(3)	(4)	(5)	(6)
Low Review Credibility	0.265*** (0.00661)	0.245*** (0.00832)	0.225*** (0.0124)	0.208*** (0.0150)	0.200*** (0.00652)	0.192*** (0.00648)
# Topics					0.0386*** (0.00216)	0.0388*** (0.00214)
Avg Rating					0.189*** (0.0127)	0.194*** (0.0126)
# Reviews					-1.75e-06*** (1.21e-07)	-1.60e-06*** (1.20e-07)
Shipped from Amazon					0.0433*** (0.00925)	0.0995*** (0.00985)
Amazon-Sold						-0.118*** (0.00759)
Constant	0.330*** (0.00372)	0.366*** (0.00501)	0.202*** (0.00958)	0.219*** (0.0120)	-0.789*** (0.0589)	-0.820*** (0.0584)
< 8 Topics Only	No	No	Yes	Yes	No	No
Excluding Amazon-Sold	No	Yes	No	Yes	No	No
Search Term $\times$ Page No. FE	No	No	No	No	Yes	Yes
Category FE	No	No	No	No	Yes	Yes
Observations	14284	9658	2528	1916	13270	13270
R^2	0.101	0.082	0.116	0.091	0.501	0.511

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.6: Product-Level Evidence, Using a RateBud Measure Excluding Review Content Quality and Constructed with RateBud's Original Weights

	(1)	(2)	(3)	(4)	(5)	(6)
Low Review Credibility	0.174*** (0.00773)	0.153*** (0.00971)	0.141*** (0.0150)	0.104*** (0.0174)	0.0505*** (0.00762)	0.0465*** (0.00753)
Amazon-Sold						-0.134*** (0.00783)
# Topics					0.0225*** (0.00217)	0.0234*** (0.00214)
Avg Rating					0.171*** (0.0132)	0.177*** (0.0131)
# Reviews					-1.92e-06*** (1.25e-07)	-1.73e-06*** (1.24e-07)
Shipped from Amazon					0.0464*** (0.00960)	0.110*** (0.0102)
Constant	0.376*** (0.00360)	0.419*** (0.00472)	0.305*** (0.00721)	0.327*** (0.00859)	-0.539*** (0.0609)	-0.583*** (0.0602)
< 8 Topics Only	No	No	Yes	Yes	No	No
Excluding Amazon-Sold	No	Yes	No	Yes	No	No
Search Term $\times$ Page No. FE	No	No	No	No	Yes	Yes
Category FE	No	No	No	No	Yes	Yes
Observations	14284	9658	2528	1916	13270	13270
R^2	0.034	0.025	0.034	0.018	0.464	0.477

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.7: Product-Level Evidence, Using a RateBud Measure Excluding Review Content Quality and Constructed with Equal Weights

	(1)	(2)	(3)	(4)	(5)	(6)
Low Review Credibility	0.176*** (0.00784)	0.165*** (0.00990)	0.155*** (0.0155)	0.124*** (0.0183)	0.0513*** (0.00791)	0.0489*** (0.00781)
Amazon-Sold						-0.135*** (0.00783)
# Topics					0.0226*** (0.00217)	0.0235*** (0.00214)
Avg Rating					0.168*** (0.0132)	0.175*** (0.0130)
# Reviews					-1.92e-06*** (1.25e-07)	-1.73e-06*** (1.24e-07)
Shipped from Amazon					0.0469*** (0.00960)	0.111*** (0.0102)
Constant	0.377*** (0.00358)	0.418*** (0.00467)	0.305*** (0.00710)	0.326*** (0.00840)	-0.527*** (0.0606)	-0.575*** (0.0600)
< 8 Topics Only	No	No	Yes	Yes	No	No
Excluding Amazon-Sold	No	Yes	No	Yes	No	No
Search Term $\times$ Page No. FE	No	No	No	No	Yes	Yes
Category FE	No	No	No	No	Yes	Yes
Observations	14284	9658	2528	1916	13270	13270
R^2	0.034	0.028	0.038	0.024	0.464	0.477

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.8: Product-Level Evidence, with Share of Time in the Buy Box

	(1)	(2)	(3)	(4)
Amazon Share	-0.153*** (0.00769)	-0.0372* (0.0183)	-0.153*** (0.00848)	-0.148*** (0.00847)
Low Review Credibility				0.203*** (0.0137)
# Topics			0.0237*** (0.00218)	0.0278*** (0.00218)
Avg Rating			0.125*** (0.0128)	0.133*** (0.0128)
# Reviews			-1.53e-06*** (1.17e-07)	-1.57e-06*** (1.18e-07)
Shipped from Amazon			0.0655*** (0.00928)	0.0619*** (0.00922)
Constant	0.447*** (0.00393)	0.327*** (0.00739)	-0.322*** (0.0586)	-0.391*** (0.0586)
< 8 Topics	No	Yes	No	No
Search Term $\times$ Page No. FE	No	No	Yes	Yes
Category FE	No	No	Yes	Yes
Observations	13617	2305	12640	12494
R^2	0.028	0.002	0.485	0.496

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.9: Product-Level Analysis of Overrepresentation of Fake Reviews, with Alternative Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Amazon-Sold	-0.135*** (0.00747)		-0.131*** (0.00744)	-0.137*** (0.00735)		-0.133*** (0.00734)	-0.137*** (0.00736)		-0.134*** (0.00735)
Low Review Credibility		0.213*** (0.0133)	0.206*** (0.0131)		0.192*** (0.0131)	0.184*** (0.0130)		0.197*** (0.0131)	0.188*** (0.0130)
# Topics	0.0220*** (0.00205)	0.0255*** (0.00208)	0.0263*** (0.00206)	0.0298*** (0.00205)	0.0322*** (0.00208)	0.0330*** (0.00205)	0.0284*** (0.00205)	0.0310*** (0.00208)	0.0318*** (0.00205)
Avg Rating	-1.375*** (0.0574)	-1.283*** (0.0584)	-1.280*** (0.0576)	-0.558*** (0.0325)	-0.535*** (0.0329)	-0.528*** (0.0325)	-0.805*** (0.0324)	-0.768*** (0.0329)	-0.763*** (0.0324)
# Reviews	-1.47e-06*** (1.17e-07)	-1.71e-06*** (1.19e-07)	-1.53e-06*** (1.18e-07)	-1.51e-06*** (1.15e-07)	-1.75e-06*** (1.18e-07)	-1.57e-06*** (1.17e-07)	-1.51e-06*** (1.15e-07)	-1.75e-06*** (1.18e-07)	-1.57e-06*** (1.17e-07)
Shipped from Amazon	0.123*** (0.00976)	0.0552*** (0.00914)	0.118*** (0.00970)	0.120*** (0.00960)	0.0518*** (0.00902)	0.115*** (0.00956)	0.120*** (0.00962)	0.0518*** (0.00904)	0.115*** (0.00958)
Share of 5-Star Reviews	3.638*** (0.114)	3.461*** (0.116)	3.465*** (0.114)	-1.928*** (0.202)	-1.776*** (0.211)	-1.823*** (0.208)	-0.659*** (0.164)	-0.527*** (0.169)	-0.571*** (0.167)
Variance of Ratings	-0.172*** (0.0124)	-0.156*** (0.0126)	-0.157*** (0.0125)						
Rating HHI				3.384*** (0.134)	3.195*** (0.141)	3.231*** (0.139)			
Rating HHI $\times$ Avg Rating							0.521*** (0.0216)	0.485*** (0.0224)	0.492*** (0.0221)
Constant	3.836*** (0.197)	3.516*** (0.200)	3.487*** (0.198)	2.067*** (0.0900)	1.954*** (0.0919)	1.917*** (0.0907)	2.845*** (0.106)	2.667*** (0.109)	2.641*** (0.107)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13420	13269	13269	13421	13270	13270	13421	13270	13270
R ²	0.516	0.515	0.527	0.532	0.528	0.541	0.530	0.526	0.539

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ **Table A.10:** Product-Level Analysis of Overrepresentation of Fake Reviews, Excluding Products with Average Ratings Below Four

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Amazon-Sold	-0.135*** (0.00698)			-0.0838*** (0.0159)			-0.139*** (0.00785)		-0.135*** (0.00782)
Low Review Credibility		0.371*** (0.0163)	0.342*** (0.0192)		0.251*** (0.0224)	0.236*** (0.0255)		0.244*** (0.0145)	0.236*** (0.0143)
# Topics							0.0246*** (0.00222)	0.0280*** (0.00224)	0.0290*** (0.00222)
Avg Rating							0.340*** (0.0165)	0.324*** (0.0166)	0.334*** (0.0164)
# Reviews							-1.64e-06*** (1.19e-07)	-1.87e-06*** (1.22e-07)	-1.69e-06*** (1.21e-07)
Shipped from Amazon							0.116*** (0.0104)	0.0475*** (0.00976)	0.112*** (0.0103)
Constant	0.448*** (0.00396)	0.389*** (0.00333)	0.431*** (0.00435)	0.341*** (0.00767)	0.297*** (0.00680)	0.316*** (0.00820)	-1.334*** (0.0766)	-1.280*** (0.0770)	-1.347*** (0.0761)
< 8 Topics Only	No	No	No	Yes	Yes	Yes	No	No	No
Excluding Amazon-Sold	No	No	Yes	No	No	Yes	No	No	No
Search Term $\times$ Page No. FE	No	No	No	No	No	No	Yes	Yes	Yes
Category FE	No	No	No	No	No	No	Yes	Yes	Yes
Observations	13700	13567	9170	2290	2281	1734	12736	12587	12587
R ²	0.027	0.037	0.033	0.012	0.052	0.047	0.496	0.496	0.509

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix C Robustness Checks for Testing H3: Distortion in Product Quality Information

C.1 Sentiment Analysis with Specialized BERT

In the main text, we use OpenAI GPT-4.1, a general-purpose model, to convert summary paragraphs into sentiment scores. Here, we use `bert-base-multilingual-uncased-sentiment`, a specialized BERT model fine-tuned to predict ratings of reviews, to convert summary paragraphs to sentiment scores measured on the scale of ratings. The results are shown in Table A.11.

C.2 Alternative Cutoffs

We repeat the sentiment analysis using the stricter D/F and more inclusive B/C/D/F definitions of low review credibility introduced in Appendix B.1. Tables A.12 and A.13 show that low-review-credibility products receive significantly more positive topic and paragraph sentiment under both definitions. The findings in the main text are therefore not sensitive to the choice of cutoff.

C.3 Excluding Review Content Quality from the RateBud Measure

Using the alternative RateBud proxies defined in Appendix B.2, we re-estimate the topic-sentiment and paragraph-sentiment specifications. Tables A.14 and A.15 show that products classified as having low review credibility still receive significantly more positive AI summaries. The product-quality-information results therefore do not depend on RateBud’s Review Content Quality criterion.

C.4 Share of Time in the Buy Box

We apply the continuous Amazon Buy Box share defined in Appendix B.3 to the topic-sentiment and paragraph-sentiment outcomes. Table A.16 reports negative coefficients on *Amazon Share* across all four specifications, consistent with the baseline findings.

C.5 Alternative Controls

We include an alternative set of controls to check the robustness of the findings in the main text. We control for additional characteristics of the rating distribution. We also include the number of extracted topics as a control variable, although its coefficient is not statistically significant in many specifications. The results reported in Tables A.17 and A.18 remain consistent with the findings in the main text.

C.6 Excluding Products with Low Average Ratings

The average ratings of most products are above four. One concern is that the results may be driven by products with low average ratings. Therefore, we exclude products with average ratings below four. Table A.19 reports the results. Column definitions follow Table 6. The results are all consistent with the findings in the main text.

C.7 Figures and Tables

Table A.11: Distortion in Product Quality Information with BERT

	(1) Paragraph	(2) Paragraph	(3) Paragraph	(4) Paragraph
Amazon-Sold	-0.0895*** (0.0114)	-0.118*** (0.0122)		
Low Review Credibility			0.173*** (0.0214)	0.167*** (0.0214)
Avg Rating	1.194*** (0.0202)	1.212*** (0.0203)	1.203*** (0.0203)	1.217*** (0.0204)
# Reviews		-9.75e-07*** (1.90e-07)		-1.22e-06*** (1.94e-07)
Shipped from Amazon		0.122*** (0.0159)		0.0647*** (0.0149)
Constant	-1.723*** (0.0898)	-1.888*** (0.0914)	-1.799*** (0.0903)	-1.905*** (0.0919)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	13516	13459	13383	13307
R^2	0.433	0.437	0.434	0.438

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.12: Distortion in Product Quality Information Using Grades D/F to Define Low Review Credibility

	(1) Topic	(2) Topic	(3) Paragraph	(4) Paragraph
Low Review Credibility	0.0606* (0.0271)	0.0593* (0.0269)	0.168*** (0.0503)	0.167*** (0.0500)
Avg Rating	0.675*** (0.00769)	0.681*** (0.00770)	1.138*** (0.0143)	1.150*** (0.0143)
# Reviews		-5.18e-07*** (7.23e-08)		-1.27e-06*** (1.34e-07)
Shipped from Amazon		0.0469*** (0.00568)		0.0441*** (0.0105)
Constant	-2.309*** (0.0342)	-2.369*** (0.0345)	-0.888*** (0.0633)	-0.969*** (0.0640)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Observations	14316	14236	14316	14236
R ²	0.491	0.498	0.465	0.470

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ **Table A.13:** Distortion in Product Quality Information Using Grades B/C/D/F to Define Low Review Credibility

	(1) Topic	(2) Topic	(3) Paragraph	(4) Paragraph
Low Review Credibility	0.0372*** (0.00426)	0.0364*** (0.00426)	0.0965*** (0.00788)	0.0946*** (0.00789)
Avg Rating	0.699*** (0.00854)	0.706*** (0.00856)	1.183*** (0.0158)	1.196*** (0.0159)
# Reviews		-3.34e-07*** (8.12e-08)		-8.45e-07*** (1.50e-07)
Shipped from Amazon		0.0473*** (0.00623)		0.0502*** (0.0115)
Constant	-2.421*** (0.0380)	-2.492*** (0.0387)	-1.110*** (0.0703)	-1.199*** (0.0717)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	13383	13307	13383	13307
R ²	0.554	0.558	0.532	0.535

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.14: Distortion in Product Quality Information, Using a RateBud Measure Excluding Review Content Quality and Constructed with RateBud's Original Weights

	(1) Topic	(2) Topic	(3) Paragraph	(4) Paragraph
Low Review Credibility	0.0145** (0.00498)	0.0144** (0.00497)	0.0343*** (0.00922)	0.0346*** (0.00922)
Avg Rating	0.695*** (0.00859)	0.703*** (0.00861)	1.173*** (0.0159)	1.187*** (0.0160)
# Reviews		-3.75e-07*** (8.13e-08)		-9.52e-07*** (1.51e-07)
Shipped from Amazon		0.0468*** (0.00625)		0.0492*** (0.0116)
Constant	-2.397*** (0.0383)	-2.469*** (0.0389)	-1.043*** (0.0709)	-1.136*** (0.0723)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	13383	13307	13383	13307
R^2	0.552	0.556	0.527	0.530

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.15: Distortion in Product Quality Information, Using a RateBud Measure Excluding Review Content Quality and Constructed with Equal Weights

	(1) Topic	(2) Topic	(3) Paragraph	(4) Paragraph
Low Review Credibility	0.0138** (0.00517)	0.0137** (0.00515)	0.0356*** (0.00958)	0.0356*** (0.00957)
Avg Rating	0.694*** (0.00857)	0.702*** (0.00859)	1.172*** (0.0159)	1.185*** (0.0159)
# Reviews		-3.75e-07*** (8.13e-08)		-9.52e-07*** (1.51e-07)
Shipped from Amazon		0.0470*** (0.00625)		0.0495*** (0.0116)
Constant	-2.393*** (0.0381)	-2.464*** (0.0388)	-1.035*** (0.0707)	-1.128*** (0.0720)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	13383	13307	13383	13307
R ²	0.552	0.556	0.527	0.530

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.16: Distortion in Product Quality Information with Share of Time in the Buy Box

	(1) Topic	(2) Topic	(3) Paragraph	(4) Paragraph
Amazon Share	-0.0725*** (0.00576)	-0.0780*** (0.00582)	-0.127*** (0.0107)	-0.131*** (0.0108)
Avg Rating	0.701*** (0.00877)	0.710*** (0.00879)	1.181*** (0.0163)	1.195*** (0.0163)
# Reviews		-2.14e-07** (8.00e-08)		-6.57e-07*** (1.49e-07)
Shipped from Amazon		0.0567*** (0.00638)		0.0665*** (0.0119)
Constant	-2.404*** (0.0389)	-2.487*** (0.0396)	-1.039*** (0.0721)	-1.148*** (0.0735)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes
Observations	12754	12675	12754	12675
R ²	0.561	0.566	0.537	0.540

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.17: Distortion in Topic Sentiment, with Alternative Controls

	(1) Topic	(2) Topic	(3) Topic	(4) Topic	(5) Topic	(6) Topic	(7) Topic	(8) Topic	(9) Topic
Amazon-Sold	-0.0672*** (0.00508)		-0.0666*** (0.00511)	-0.0674*** (0.00509)		-0.0669*** (0.00513)	-0.0673*** (0.00509)		-0.0667*** (0.00513)
Low Review Credibility		0.0434*** (0.00906)	0.0396*** (0.00900)		0.0362*** (0.00912)	0.0321*** (0.00906)		0.0383*** (0.00911)	0.0342*** (0.00905)
Avg Rating	0.836*** (0.0391)	0.868*** (0.0399)	0.869*** (0.0396)	0.595*** (0.0225)	0.592*** (0.0229)	0.595*** (0.0227)	0.586*** (0.0225)	0.592*** (0.0228)	0.595*** (0.0226)
# Reviews	-2.08e-07** (7.91e-08)	-3.26e-07*** (8.12e-08)	-2.36e-07** (8.09e-08)	-2.14e-07** (7.93e-08)	-3.32e-07*** (8.14e-08)	-2.42e-07** (8.12e-08)	-2.12e-07** (7.93e-08)	-3.29e-07*** (8.14e-08)	-2.39e-07** (8.12e-08)
Shipped from Amazon	0.0821*** (0.00664)	0.0494*** (0.00624)	0.0811*** (0.00666)	0.0817*** (0.00665)	0.0491*** (0.00626)	0.0810*** (0.00668)	0.0819*** (0.00666)	0.0494*** (0.00626)	0.0811*** (0.00668)
# Topics	-0.00482*** (0.00140)	-0.00416** (0.00142)	-0.00376** (0.00141)	-0.00337* (0.00142)	-0.00321* (0.00144)	-0.00277 (0.00143)	-0.00393** (0.00142)	-0.00365* (0.00144)	-0.00321* (0.00143)
Share of 5-Star Reviews	0.00878 (0.0776)	-0.0367 (0.0790)	-0.0346 (0.0784)	0.0698 (0.140)	0.243 (0.147)	0.218 (0.146)	0.332** (0.113)	0.460*** (0.117)	0.438*** (0.116)
Variance of Ratings	0.0657*** (0.00847)	0.0734*** (0.00864)	0.0727*** (0.00858)						
Rating HHI				0.212* (0.0931)	0.0802 (0.0976)	0.0986 (0.0969)			
Rating HHI $\times$ Avg Rating							0.00369 (0.0149)	-0.0172 (0.0155)	-0.0136 (0.0154)
Constant	-3.129*** (0.134)	-3.245*** (0.137)	-3.259*** (0.136)	-2.150*** (0.0624)	-2.180*** (0.0639)	-2.199*** (0.0634)	-2.185*** (0.0733)	-2.249*** (0.0755)	-2.261*** (0.0750)
Search Term $\times$ Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13458	13306	13306	13459	13307	13307	13459	13307	13307
R ²	0.564	0.560	0.566	0.562	0.558	0.564	0.561	0.558	0.564

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.18: Distortion in Paragraph Sentiment, with Alternative Controls

	(1) Paragraph	(2) Paragraph	(3) Paragraph	(4) Paragraph	(5) Paragraph	(6) Paragraph	(7) Paragraph	(8) Paragraph	(9) Paragraph
Amazon-Sold	-0.110*** (0.00946)		-0.109*** (0.00950)	-0.111*** (0.00944)		-0.111*** (0.00949)	-0.111*** (0.00945)		-0.110*** (0.00950)
Low Review Credibility		0.107*** (0.0168)	0.100*** (0.0167)		0.0886*** (0.0169)	0.0818*** (0.0168)		0.0928*** (0.0169)	0.0860*** (0.0168)
Avg Rating	1.045*** (0.0728)	1.121*** (0.0740)	1.122*** (0.0736)	0.911*** (0.0418)	0.913*** (0.0423)	0.920*** (0.0421)	0.818*** (0.0417)	0.838*** (0.0422)	0.842*** (0.0420)
# Reviews	-6.31e-07*** (1.47e-07)	-8.43e-07*** (1.51e-07)	-6.97e-07*** (1.50e-07)	-6.54e-07*** (1.47e-07)	-8.68e-07*** (1.51e-07)	-7.19e-07*** (1.50e-07)	-6.51e-07*** (1.47e-07)	-8.64e-07*** (1.51e-07)	-7.16e-07*** (1.50e-07)
Shipped from Amazon	0.109*** (0.0124)	0.0543*** (0.0116)	0.106*** (0.0124)	0.107*** (0.0123)	0.0527*** (0.0116)	0.105*** (0.0124)	0.107*** (0.0124)	0.0530*** (0.0116)	0.106*** (0.0124)
# Topics	-0.00611* (0.00260)	-0.00417 (0.00264)	-0.00351 (0.00262)	-0.00127 (0.00264)	-0.000409 (0.00267)	0.000314 (0.00265)	-0.00246 (0.00263)	-0.00140 (0.00266)	-0.000673 (0.00265)
Share of 5-Star Reviews	0.765*** (0.145)	0.649*** (0.147)	0.653*** (0.146)	-0.884*** (0.260)	-0.590* (0.271)	-0.631* (0.270)	-0.112 (0.211)	0.111 (0.217)	0.0743 (0.216)
Variance of Ratings	0.0536*** (0.0158)	0.0704*** (0.0160)	0.0693*** (0.0159)						
Rating HHI				1.396*** (0.173)	1.155*** (0.181)	1.186*** (0.180)			
Rating HHI × Avg Rating							0.176*** (0.0277)	0.136*** (0.0287)	0.142*** (0.0285)
Constant	-1.117*** (0.249)	-1.397*** (0.254)	-1.420*** (0.253)	-0.0808 (0.116)	-0.162 (0.118)	-0.193 (0.117)	0.126 (0.136)	-0.0211 (0.140)	-0.0420 (0.139)
Search Term × Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	13458	13306	13306	13459	13307	13307	13459	13307	13307
R ²	0.536	0.535	0.540	0.538	0.535	0.541	0.537	0.535	0.540

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ **Table A.19: Distortion in Product Quality Information, Excluding Products with Average Ratings Below Four**

	Main Specifications				Robustness Checks			
	(1) Topic	(2) Paragraph	(3) Topic	(4) Paragraph	(5) Topic	(6) Paragraph	(7) Topic	(8) Paragraph
Amazon-Sold	-0.0652*** (0.00503)	-0.112*** (0.00951)			-0.0650*** (0.00506)	-0.112*** (0.00956)		
Low Review Credibility			0.0353*** (0.00927)	0.0922*** (0.0175)	0.0312*** (0.00921)	0.0850*** (0.0174)	0.0270** (0.0104)	0.0772*** (0.0197)
Avg Rating	0.721*** (0.0106)	1.290*** (0.0200)	0.715*** (0.0107)	1.279*** (0.0202)	0.720*** (0.0106)	1.288*** (0.0201)	0.674*** (0.0128)	1.215*** (0.0242)
# Reviews	-2.41e-07** (7.61e-08)	-7.15e-07*** (1.44e-07)	-3.52e-07*** (7.82e-08)	-9.20e-07*** (1.47e-07)	-2.65e-07*** (7.79e-08)	-7.71e-07*** (1.47e-07)	-5.19e-07*** (1.22e-07)	-1.11e-06*** (2.31e-07)
Shipped from Amazon	0.0780*** (0.00666)	0.107*** (0.0126)	0.0465*** (0.00626)	0.0523*** (0.0118)	0.0776*** (0.00667)	0.106*** (0.0126)	0.0720*** (0.00752)	0.0991*** (0.0142)
Constant	-2.554*** (0.0478)	-1.610*** (0.0905)	-2.523*** (0.0483)	-1.553*** (0.0911)	-2.552*** (0.0480)	-1.603*** (0.0907)	-2.338*** (0.0577)	-1.269*** (0.109)
Search Term × Page No. FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Category FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12775	12775	12625	12625	12625	12625	8601	8601
R ²	0.495	0.486	0.489	0.483	0.497	0.489	0.507	0.499

Standard errors are in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix D Robustness Checks for Testing H4: Distortion in Market Outcomes

D.1 Estimated Coefficients

Table A.20 reports the estimated coefficients plotted in Figures 3 and 4 and their standard errors.

D.2 Alternative Introduction Dates

The baseline sales-rank analysis uses June 12, 2023, as the introduction date, because major media outlets reported on that date that Amazon was already using generative AI to summarize product reviews. Because Amazon formally announced the feature on August 14, 2023 (Schmerhorn, 2023), and the rollout of AI summarization was gradual, the exact product-level introduction date may be measured with noise. To address this concern, we rerun the event-study specification in Equation (3) using August 14, 2023, as the alternative introduction date.

Figures A.1 and A.2 report the results. The estimates are qualitatively similar to the baseline results in Figures 3 and 4. Before the alternative introduction date, there is no evidence of differential pre-trends between products with higher and lower exposure to fake reviews. After the alternative introduction date, the coefficients move in the same direction as in the baseline analysis: Amazon-sold products experience a relative worsening in sales ranks, while low-review-credibility products experience a relative improvement in sales ranks. These results suggest that the market-outcome evidence is not driven by the specific date used to anchor the event study.

D.3 Results in the Full Sample

As a robustness check, we re-estimate Equation (3) in the full sample. Figures A.3 and A.4 report the results using Amazon-sold status and low review credibility, respectively. The results closely mirror those from the sample with AI summaries.

D.4 Figures and Tables

Table A.20: Estimated Regression Coefficients

	(1) Amazon-Sold	(2) Low Review Credibility
-16	-0.025 (0.074)	-0.151 (0.187)
-15	-0.043 (0.071)	-0.195 (0.179)
-14	-0.006 (0.065)	-0.117 (0.178)
-13	0.019 (0.063)	-0.018 (0.176)
-12	0.081 (0.060)	0.024 (0.180)
-11	0.047 (0.058)	-0.005 (0.176)
-10	0.019 (0.057)	-0.041 (0.165)
-9	0.047 (0.058)	-0.054 (0.151)
-8	0.043 (0.058)	0.041 (0.136)
-7	0.092 (0.057)	0.085 (0.130)
-6	0.075 (0.055)	-0.105 (0.129)
-5	0.032 (0.054)	-0.039 (0.114)
-4	0.086* (0.050)	-0.098 (0.113)
-3	0.045 (0.039)	-0.167* (0.089)
-2	0.029 (0.026)	-0.012 (0.053)
0	-0.017 (0.023)	-0.041 (0.063)
1	0.033 (0.032)	-0.019 (0.065)
2	0.040 (0.038)	-0.068 (0.087)
3	0.076* (0.044)	-0.046 (0.102)
4	0.084* (0.049)	-0.156 (0.120)
5	0.105* (0.055)	-0.129 (0.128)
6	0.135** (0.059)	-0.228* (0.130)
7	0.155*** (0.060)	-0.293** (0.127)
8	0.148** (0.059)	-0.247* (0.126)
9	0.108* (0.058)	-0.258** (0.125)
10	0.122** (0.057)	-0.345** (0.135)
11	0.141** (0.058)	-0.356*** (0.130)
12	0.154*** (0.058)	-0.319** (0.130)
13	0.138** (0.058)	-0.261** (0.130)
14	0.178*** (0.060)	-0.227 (0.138)
15	0.193*** (0.062)	-0.312** (0.151)
16	0.196*** (0.065)	-0.232 (0.146)
17	0.228*** (0.067)	-0.306** (0.155)
18	0.210*** (0.067)	-0.299* (0.175)
Observations	4608082	1422527
R ²	0.816	0.820

Note. Event time is measured in consecutive 30-day periods. Month -1 is the omitted reference period. All specifications include product and date fixed effects. Standard errors are clustered at the product level. Standard errors are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A.1: Comparison between Amazon-Sold and Non-Amazon-Sold Products with August 14, 2023, as the Introduction Date

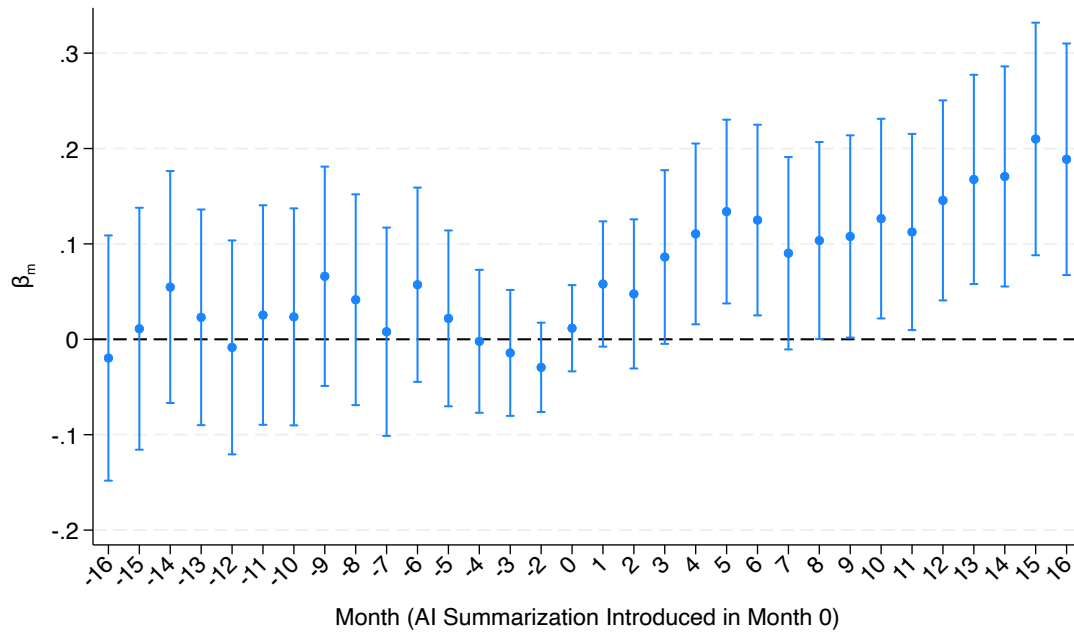


Figure A.2: Comparison between Low-Review-Credibility Products and Other Products with August 14, 2023, as the Introduction Date

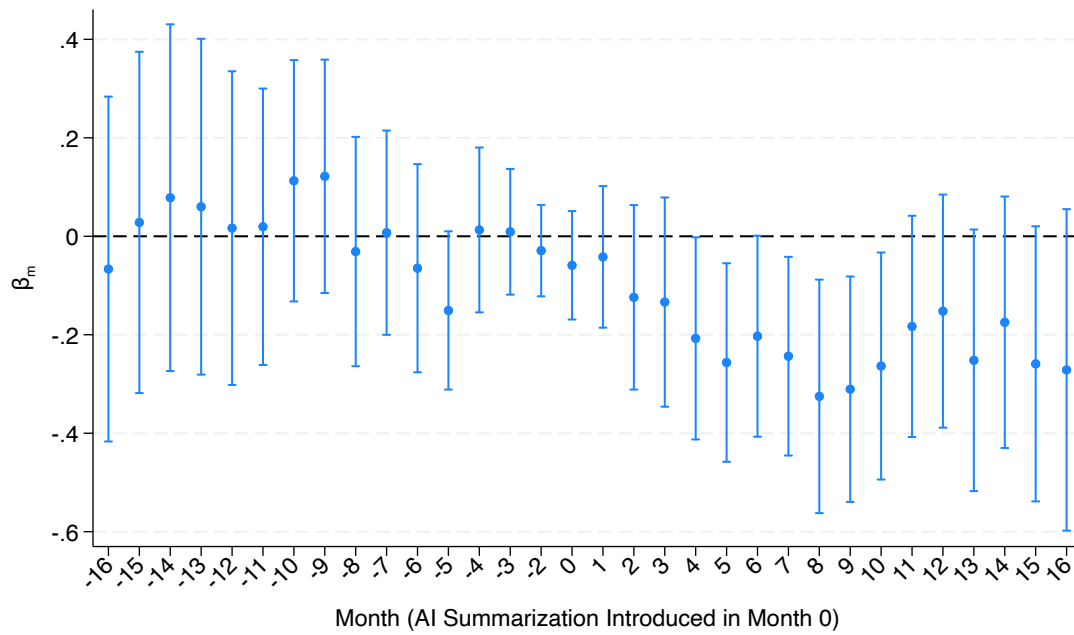


Figure A.3: Comparison between Amazon-Sold and Non-Amazon-Sold Products in the Full Sample

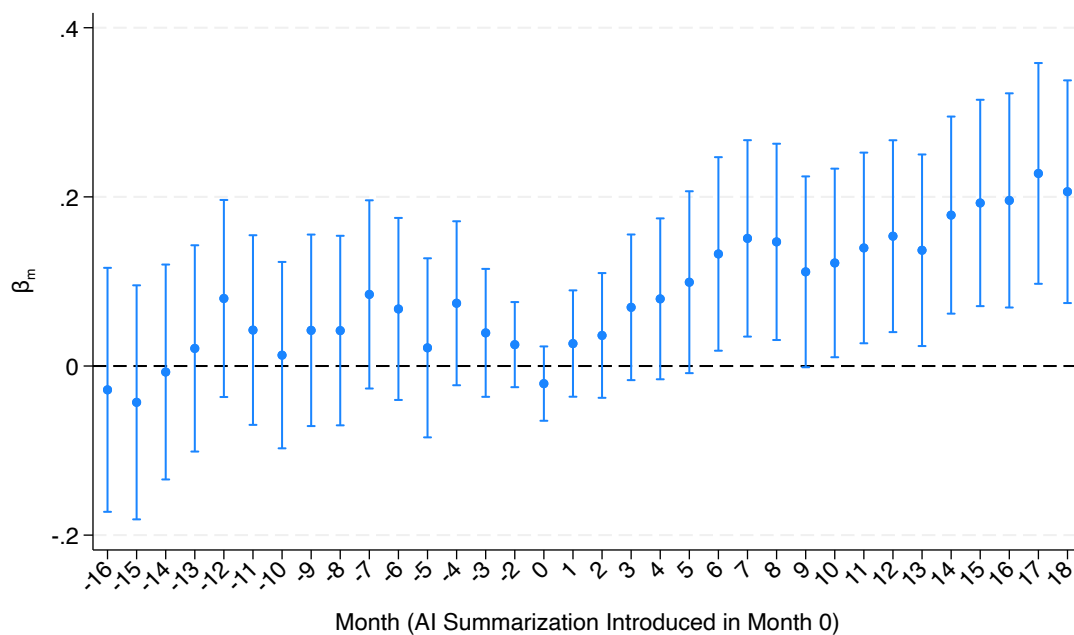
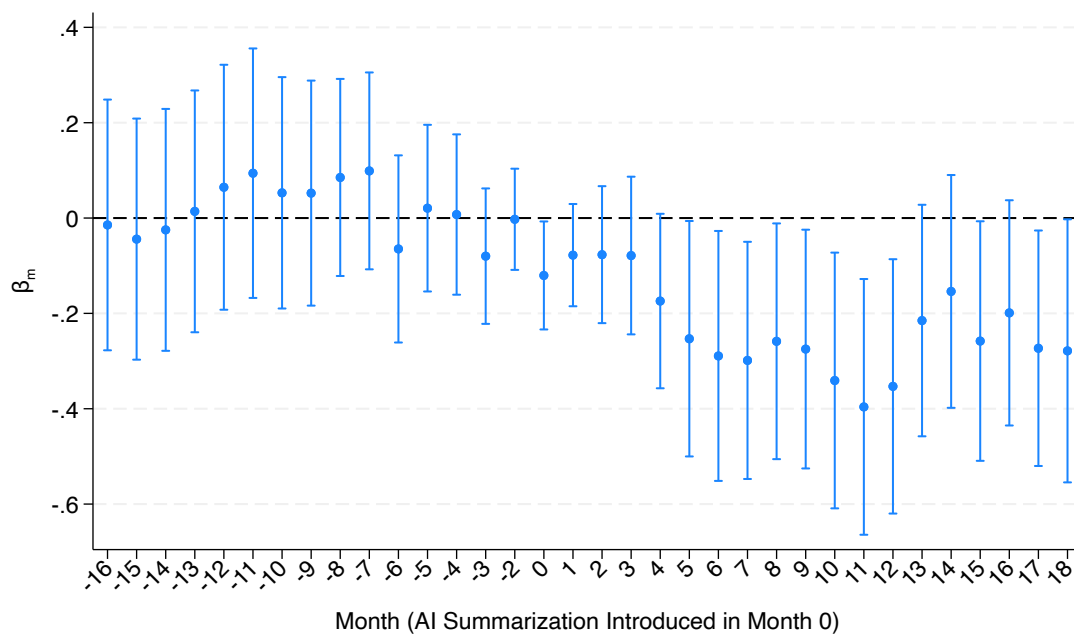


Figure A.4: Comparison between Low-Review-Credibility Products and Other Products in the Full Sample



Appendix E Additional Summary Statistics

This appendix reports additional summary statistics for Data Sources 3 and 4. Table A.21 reports RateBud review-credibility grades for the full sample, and Table A.22 reports product-level characteristics of the Keepa sample used to test Hypothesis 4. To construct Table A.22, we first identify the 6,823 products retained in the Keepa sample and then use the corresponding observations in Data Source 2 to calculate summary statistics for their Amazon product characteristics. Compared with the 14,479 products with AI summaries in Data Source 2, products in the Keepa sample tend to have more reviews, lower prices, and a slightly higher Amazon-sold share. Regarding prices, the 25th percentile and median are similar across these two sources. Nevertheless, the two samples have very similar rating distributions, with the same median and interquartile range.

Table A.21: Additional Summary Statistics for Data Source 3 (RateBud): The Full Sample

Grade	Meaning	All		Amazon-Sold		Non-Amazon-Sold	
		Count	Percentage	Count	Percentage	Count	Percentage
A	Highly trustworthy	9,998	60.07%	3,667	73.08%	6,314	54.49%
B	Generally reliable	4,777	28.70%	1,049	20.90%	3,717	32.08%
C	Mixed signals	1,306	7.85%	224	4.46%	1,076	9.29%
D	Significant concerns	276	1.66%	39	0.78%	235	2.03%
F	High risk of manipulation	287	1.72%	39	0.78%	245	2.11%
Total		16,644	100.00%	5,018	100.00%	11,587	100.00%

Note. RateBud assigns review-credibility grades A, B, C, D, or F, with lower grades indicating lower credibility. Percentages are calculated within columns. The *All* column includes the 16,644 products with available RateBud grades among the 16,851 products in Data Source 2. The *Amazon-Sold* and *Non-Amazon-Sold* columns sum to 16,605, rather than 16,644, because Amazon-sold status is missing for 39 out-of-stock products; these products remain in the *All* column.

Table A.22: Additional Summary Statistics for Data Source 4 (Keepa)

Statistic	Mean	SD	25th pct.	Median	75th pct.
<i>Product-level summary statistics of the 6,823 products in Data Source 4</i>					
# Reviews	12,604.76	32,163.09	1,106.00	3,677	11,913
Average Rating	4.455	0.260	4.3	4.5	4.6
Price	90.59	191.96	17.99	33.08	84.98
Percentage of Amazon-Sold Products	37.81%				

Appendix F Exploratory Evidence on the Mechanism

Having tested the overrepresentation of fake reviews, we next explore why fake reviews mention more of the topics extracted by AI summaries, i.e., $\mathbb{E}[W_{ij}|F_{ij} = 1] > \mathbb{E}[W_{ij}|F_{ij} = 0]$. We test the two channels proposed in Section 4.2.

F.1 Common-Topic Alignment Channel

The common-topic alignment channel posits that fake reviews draw on a shared pool of generic product merits that coincide with the recurring topics extracted by AI summaries. It therefore predicts that the fake-authentic difference $\mathbb{E}[W_{ij} \mid F_{ij} = 1] - \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ in topic mentions should be mainly attributable to recurring and broadly applicable topics rather than idiosyncratic ones. Here, we decompose the fake-authentic difference and test this prediction.

We first use the lexical-or-semantic matching procedure introduced in Section 5.1.1 to identify the extracted topics mentioned by each review. Because each review is associated with a unique product, we suppress the product subscript in this appendix. Let $j(i)$ denote the product associated with review i , and define $F_i \equiv F_{i,j(i)}$ and $W_i \equiv W_{i,j(i)}$.

Let $\mathcal{R}$ denote the set of reviews in the review-level sample, and let $\mathcal{J}$ denote the corresponding set of products. Define $\mathcal{R}^F \equiv \{i \in \mathcal{R} : F_i = 1\}$ and $\mathcal{R}^A \equiv \{i \in \mathcal{R} : F_i = 0\}$ as the sets of fake and authentic reviews, respectively. Let $\mathcal{K}_j$ denote the set of common topics extracted by the AI summarization algorithm for product j . Define $\mathcal{K} \equiv \bigcup_{j \in \mathcal{J}} \mathcal{K}_j$ as the set of topic labels appearing in at least one product’s extracted topic set.

For each review $i \in \mathcal{R}$ and topic $k \in \mathcal{K}$, define $M_{ik} \equiv 1\{k \in \mathcal{K}_{j(i)}\}1\{\text{review } i \text{ mentions topic } k\}$. By construction,

$$W_i = \sum_{k \in \mathcal{K}} M_{ik} = \sum_{k \in \mathcal{K}_{j(i)}} 1\{\text{review } i \text{ mentions topic } k\}.$$

We construct three topic-level mention-rate measures. The *fake-review mention rate* is

$$\text{FMR}_k \equiv \frac{1}{|\mathcal{R}^F|} \sum_{i \in \mathcal{R}^F} M_{ik},$$

and the *authentic-review mention rate* is

$$\text{AMR}_k \equiv \frac{1}{|\mathcal{R}^A|} \sum_{i \in \mathcal{R}^A} M_{ik}.$$

The *overall mention rate* is

$$\text{OMR}_k \equiv \frac{1}{|\mathcal{R}|} \sum_{i \in \mathcal{R}} M_{ik}.$$

These measures capture how frequently topic k is mentioned among fake reviews, authentic reviews, or all reviews, respectively.

We next decompose the aggregate fake-authentic difference reported in the main text as evidence of the overrepresentation of fake reviews. Equation (4) decomposes the sample analog of $\mathbb{E}[W_{ij} \mid$

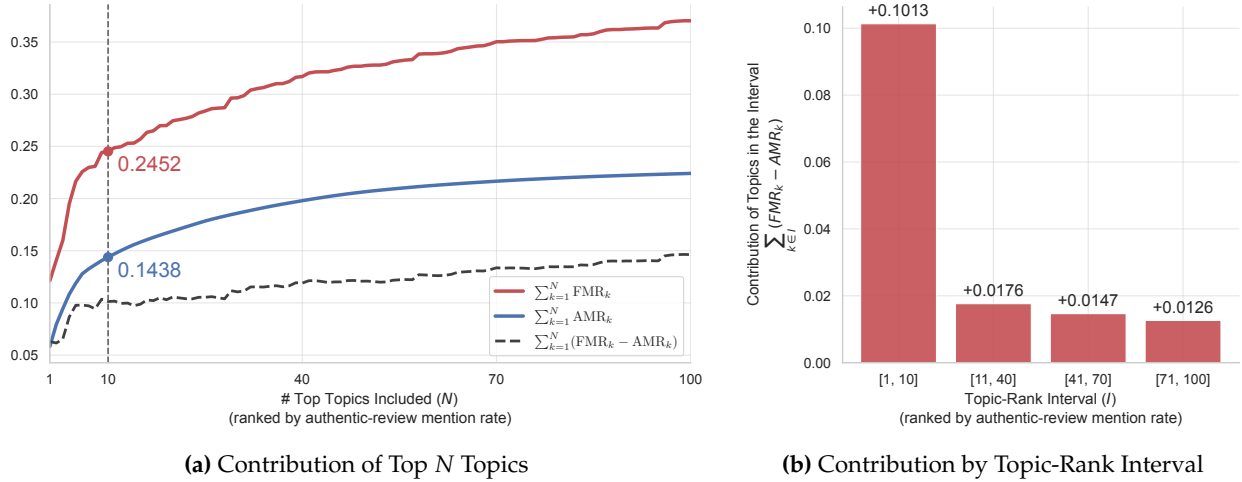
$F_{ij} = 1] - \mathbb{E}[W_{ij} \mid F_{ij} = 0]$ into additive topic-level contributions:

$$\begin{aligned}
\frac{1}{|\mathcal{R}^F|} \sum_{i \in \mathcal{R}^F} W_i - \frac{1}{|\mathcal{R}^A|} \sum_{i \in \mathcal{R}^A} W_i &= \frac{1}{|\mathcal{R}^F|} \sum_{i \in \mathcal{R}^F} \sum_{k \in \mathcal{K}} M_{ik} - \frac{1}{|\mathcal{R}^A|} \sum_{i \in \mathcal{R}^A} \sum_{k \in \mathcal{K}} M_{ik} \\
&= \sum_{k \in \mathcal{K}} \left[\frac{1}{|\mathcal{R}^F|} \sum_{i \in \mathcal{R}^F} M_{ik} - \frac{1}{|\mathcal{R}^A|} \sum_{i \in \mathcal{R}^A} M_{ik} \right] \\
&= \sum_{k \in \mathcal{K}} (\text{FMR}_k - \text{AMR}_k).
\end{aligned} \tag{4}$$

Thus, $\text{FMR}_k - \text{AMR}_k$ is topic k 's contribution to the aggregate fake-authentic difference.

In practice, we rank all topics in descending order of their authentic-review mention rates. Ranking topics exclusively by their frequency among authentic reviews ensures that their classification as popular is not mechanically driven by their prevalence among fake reviews. For each value of N , we calculate the number of the top N topics mentioned per fake and per authentic review, i.e., $\sum_{k=1}^N \text{FMR}_k$ and $\sum_{k=1}^N \text{AMR}_k$. The difference between them is the contribution of the top N topics to the aggregate fake-authentic difference, i.e., $\sum_{k=1}^N (\text{FMR}_k - \text{AMR}_k)$. We examine how this cumulative contribution changes as progressively lower-ranked topics are added, i.e., how $\sum_{k=1}^N (\text{FMR}_k - \text{AMR}_k)$ changes as N increases. This procedure identifies whether the contribution to the aggregate difference is concentrated in the most popular topics among authentic reviews or distributed evenly across topics.

Figure A.5: Decomposition of the Fake-Authentic Difference by Topics



Note. Topics are indexed in descending order of their authentic-review mention rates, so that $\text{AMR}_1 \geq \text{AMR}_2 \geq \dots$. For each N , Panel (a) plots $\sum_{k=1}^N \text{FMR}_k$ and $\sum_{k=1}^N \text{AMR}_k$, the number of the top N topics mentioned per fake and per authentic review, respectively. The dashed line plots $\sum_{k=1}^N (\text{FMR}_k - \text{AMR}_k)$, the contribution of the top N topics to the aggregate fake-authentic difference. Panel (b) groups topic ranks into intervals I and reports $\sum_{k \in I} (\text{FMR}_k - \text{AMR}_k)$, the contribution of topics in each interval to the fake-authentic difference. The first interval, $I = [1, 10]$, accounts for approximately 69% of $\sum_{k=1}^{100} (\text{FMR}_k - \text{AMR}_k)$.

Figure A.5 shows that the aggregate fake-authentic difference emerges primarily among the topics that are most frequently mentioned by authentic reviewers. As shown in Panel (a), the ten highest-ranked topics are mentioned 24.52 times per 100 fake reviews, compared with 14.38 times

per 100 authentic reviews. Thus, fake reviews mention these ten topics 70.5% more frequently, producing a difference of 10.14 topic mentions per 100 reviews. The dashed line plots the cumulative contribution of the top N topics, i.e., $\sum_{k=1}^N (\text{FMR}_k - \text{AMR}_k)$. It rises sharply as the first ten topics are added but increases much more gradually thereafter, indicating that the overrepresentation of fake reviews is driven disproportionately by the most popular topics among authentic reviews.

Interestingly, the $\sum_{k=1}^N \text{FMR}_k$ curve remains above the $\sum_{k=1}^N \text{AMR}_k$ curve at every cutoff from $N = 1$ through $N = 100$. Because the ranking is constructed exclusively from authentic-review mention rates, this cumulative dominance is not mechanical. Instead, it suggests that regardless of where the cutoff is placed, fake reviews mention more of the topics that are most frequently mentioned by authentic reviewers than authentic reviews themselves.

Panel (b) shows this concentration more explicitly by reporting the contributions of topics in different rank intervals. The ten highest-ranked topics account for approximately 69% of the fake-authentic difference generated by the top 100 topics, and 60% of the total difference ($0.40 - 0.23$). Topics ranked below the top ten make considerably smaller contributions. Therefore, the overrepresentation of fake reviews is driven disproportionately by the top ten most popular topics among authentic reviewers.

We then examine what these top ten most popular topics among authentic reviews are. Table A.23 lists them directly. We find that fake reviews are substantially more likely than authentic reviews to mention the ten most frequently mentioned topics in the authentic-review corpus, including “quality” (12.13% versus 5.82%), “sound quality” (3.52% versus 1.39%), “battery life” (2.16% versus 1.05%), and “brightness” (1.32% versus 0.39%). This is consistent with the previous finding that fake reviews mention more of the topics that are most frequently mentioned by authentic reviewers than authentic reviews themselves.

To measure whether a topic represents a broadly applicable product attribute, we calculate its *product breadth*. Let $\mathcal{K}_j$ denote the set of extracted topics for product j . Product breadth is defined as $\text{PB}_k \equiv \sum_{j \in \mathcal{J}} \frac{1\{k \in \mathcal{K}_j\}}{|\mathcal{J}|}$, the share of products whose extracted topic set contains topic k . A higher value indicates that the same topic recurs across a broader range of products and is therefore more broadly applicable. “Quality” appears in the extracted topic sets of 74.48% of products. Other recurring topics also span substantial portions of the sample, including “size” (18.38%), “fit” (14.58%), “battery life” (13.95%), “comfort” (13.00%), “sound quality” (7.77%), and “brightness” (7.45%). These topics capture general dimensions along which consumers commonly evaluate products rather than highly specific usage experiences. This result provides evidence that many of the topics driving the aggregate fake-authentic difference represent broadly applicable product merits.

Taken together, the decomposition shows that the contribution to the aggregate fake-authentic difference is concentrated in a small number of topics that are most frequently discussed by authentic reviewers, many of which are broadly applicable product merits. This pattern provides exploratory evidence consistent with the common-topic alignment channel: fake reviews disproportionately emphasize topics that recur most frequently among authentic reviews and are

Table A.23: Fake Reviews and Topics Popular among Authentic Reviewers

Topic	Fake-Review Mention Rate	Authentic-Review Mention Rate	Overall Mention Rate	Product Breadth (%)
Quality	0.1213	0.0582	0.0748	74.48
Fit	0.0197	0.0212	0.0208	14.58
Size	0.0188	0.0149	0.0160	18.38
Sound quality	0.0352	0.0139	0.0195	7.77
Battery life	0.0216	0.0105	0.0134	13.95
Taste	0.0092	0.0091	0.0091	2.22
Comfort	0.0041	0.0047	0.0046	13.00
Magnification	0.0011	0.0040	0.0032	0.32
Brightness	0.0132	0.0039	0.0064	7.45
Privacy	0.0010	0.0033	0.0027	0.32

Note. The table reports the ten topics with the highest authentic-review mention rates. The fake-review, authentic-review, and overall mention rates equal the numbers of reviews in the corresponding samples that mention the topic, divided by the total numbers of reviews in those samples. Product breadth is the percentage of products whose extracted topic set $\mathcal{K}_j$ contains the corresponding topic.

broadly applicable to many products. Additionally, fake reviews mention more of these topics than authentic reviews themselves. Because AI summarization algorithms are designed to extract these recurring topics, this concentration makes the content of fake reviews particularly likely to overlap with the topics extracted by AI summaries.

F.2 Review-Length Channel

We next examine the complementary review-length channel. Table 2 shows that fake reviews are significantly longer than authentic reviews, while Table 4 shows that fake reviews mention more extracted topics. We examine whether the greater length of fake reviews corresponds to broader topic coverage or merely to additional sentences that repeat the same topic.

We divide each review into sentences, convert the sentences into embeddings, use UMAP to reduce the dimensionality of the embeddings, and apply HDBSCAN to cluster semantically related sentences. We count the number of sentences assigned to topic clusters for each review, and the number of unique topic clusters represented in each review.

Table A.24: Review Length and Topic Breadth

	Fake Reviews	Authentic Reviews	Authentic 5-Star Reviews
# Sentences that Mention Topics	2.3404	1.4847 ($p < 0.001$)	1.5767 ($p < 0.001$)
# Unique Topics Mentioned by Each Review	1.2929	0.9354 ($p < 0.001$)	0.9844 ($p < 0.001$)

Table A.24 reports the results. Fake reviews contain more sentences assigned to topic clusters and mention more unique topics than either all authentic reviews or authentic five-star reviews. On average, fake reviews mention 1.2929 unique topics, compared with 0.9354 among all authentic reviews and 0.9844 among authentic five-star reviews. Thus, the additional content in fake reviews does not merely repeat a single topic; fake reviews cover a broader range of distinct topics.

Taken together, the results are consistent with the two complementary channels described in Section 4.2. Fake reviews are concentrated on generic topics that are popular in the review corpus, which are topics that AI summarization algorithms are designed to extract. Greater review length expands the number of distinct common topics that an individual fake review can cover.

Appendix G Relationship between Review-Level and Product-Level Overrepresentation

Hypothesis 1 (H1) is defined at the review level, whereas Hypothesis 2 (H2) concerns an association across products. This appendix clarifies the relationship between the two hypotheses.

Draw a review at random from the population of reviews, and let j denote its product. Define

$$f_j \equiv \mathbb{E}[F_{ij} \mid j], \quad w_j \equiv \mathbb{E}[W_{ij} \mid j],$$

and

$$w_{F,j} \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 1, j], \quad w_{A,j} \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 0, j], \quad \delta_j \equiv w_{F,j} - w_{A,j}.$$

Let $p \equiv \mathbb{E}[F_{ij}]$. The review-level difference in H1 is

$$D_1 \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 1] - \mathbb{E}[W_{ij} \mid F_{ij} = 0].$$

Because F_{ij} is binary, $\text{Cov}(W_{ij}, F_{ij}) = p(1 - p)D_1$. Thus, H1 is equivalent to $\text{Cov}(W_{ij}, F_{ij}) > 0$.

Now, let us consider H2. Suppressing product-level controls for exposition, the population analogue of H2 is a positive correlation between w_j and f_j , or $\text{Cov}(w_j, f_j) > 0$. With controls, the same interpretation applies to the corresponding conditional or residualized correlation.

The law of total covariance leads to Equation (5), which shows why H1 and H2 are not nested without additional assumptions.

$$\underbrace{p(1 - p)D_1}_{\text{H1: total review-level association}} = \underbrace{\mathbb{E}[f_j(1 - f_j)\delta_j]}_{\text{within-product component}} + \underbrace{\text{Cov}(w_j, f_j)}_{\text{between-product component associated with H2}}. \quad (5)$$

To see this, note that

$$\text{Cov}(W_{ij}, F_{ij} \mid j) = f_j(1 - f_j)(w_{F,j} - w_{A,j}) = f_j(1 - f_j)\delta_j,$$

and apply $\text{Cov}(W_{ij}, F_{ij}) = \mathbb{E}[\text{Cov}(W_{ij}, F_{ij} \mid j)] + \text{Cov}(\mathbb{E}[W_{ij} \mid j], \mathbb{E}[F_{ij} \mid j])$.

Under the mean independence condition used in Proposition 1, H1 and H2 are equivalent.

Proposition 1. Suppose that $\mathbb{E}[W_{ij} \mid F_{ij}, j] = \mathbb{E}[W_{ij} \mid F_{ij}]$ and $\mathbb{V}(f_j) > 0$. Then H1 holds if and only if $\text{Cov}(w_j, f_j) > 0$. Formally, for any two product groups J_1 and J_2 satisfying $\mathbb{E}[f_j \mid j \in J_1] > \mathbb{E}[f_j \mid j \in J_2]$, H1 holds if and only if $\mathbb{E}[w_j \mid j \in J_1] > \mathbb{E}[w_j \mid j \in J_2]$.

Proof. Let $\mu_F \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 1]$ and $\mu_A \equiv \mathbb{E}[W_{ij} \mid F_{ij} = 0]$. Under the mean independence condition,

$$w_j = \mu_A(1 - f_j) + \mu_F f_j = \mu_A + (\mu_F - \mu_A)f_j.$$

Therefore, $\text{Cov}(w_j, f_j) = (\mu_F - \mu_A)\mathbb{V}(f_j)$, and the difference in mean w_j between J_1 and J_2 equals $(\mu_F - \mu_A)\{\mathbb{E}[f_j \mid j \in J_1] - \mathbb{E}[f_j \mid j \in J_2]\}$. Both have the same sign as $D_1 = \mu_F - \mu_A$. $\square$

The mean independence condition is sufficient for equivalence, but it is not imposed in the empirical analysis. Accordingly, the review-level and product-level analyses should be interpreted as complementary evidence.